The AI Agentic Penalty in Interpersonal Trust*

Paul M. Gorny, Emike Nasamu, Mengjie Wang

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Abstract

Handing an AI the initiative may make people listen to it less. We study whether AI acting on people's behalf changes their reliance on its knowledge when deciding whether to trust others. In a preregistered, incentivised trust-game experiment with 1,596 participants, a machine-learning classifier assessed whether each matched trustee would return money. Holding the assessment and actions fixed, we randomly assigned the AI to (i) predict the trustee's behaviour, (ii) recommend a choice, or (iii) preset it for participants to confirm or override at no cost. Participants relied on the AI assistance: prediction and advice alike increased the proportion of choices consistent with the AI's assessment by about 34 percentage points over a baseline without the AI. When the AI took the initiative, however, reliance fell by 8–9 points, about a quarter of this influence. This *AI agentic penalty* is symmetric, weakening both encouragements to trust and warnings, contrary to status-quo bias, and not explained by perceived AI capabilities or trait reactance. Instead, beliefs move less with the assessment, and the revealed reluctance matches baseline participants' stated preference against delegation, consistent with a preference-based explanation. Retaining final authority is therefore not enough: people respond to who initiates the action.

Keywords: agentic AI, algorithmic advice, delegation, decision rights, defaults, trust
JEL Classifications: C91, D83, D91, O33

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1 Introduction

If decision support systems take charge, humans could stop listening. A useful characterisation of an artificial intelligence (AI) system is that it produces a prediction (Agrawal et al., 2018). Such a *prediction* generated by AI can inform a decision, or allow anticipating its consequences, or *recommend* or *enact* a choice. For an investor, the distinction is between an algorithm that assesses the prospects of an investment, recommends a portfolio, or executes a trade (Zhang et al., 2021). For an employer, it is between assessing a candidate's future performance, recommending whom to advance, or advancing candidates subject to human review. These arrangements allocate different roles to the AI even when the underlying information is unchanged and a person retains the final authority. Human oversight of algorithms commonly takes this form: the system proposes, and a person confirms or overrides, as when judges decide whether to follow a risk-assessment algorithm's bail recommendation (Angelova et al., 2026).[1] Each of these arrangements automates a further step of the same decision process. What varies across arrangements is whether the system stops at the prediction, additionally supplies the judgement that turns the prediction into an action, or takes the action itself (Onnasch et al., 2014). Their economic value therefore depends on more than the quality of the prediction. It also depends on how assigning the AI a role in the decision process changes people's willingness to use the AI and therefore its underlying knowledge.

How an AI's role shapes the use of its knowledge is particularly crucial when the decision concerns whether to trust another person. In lending, hiring, and many other exchanges, one party must decide whether to rely on another's willingness to honour an obligation. AI systems already assist in such decisions at scale: before a person decides whether to proceed, algorithms predict whether borrowers will repay (Fuster et al., 2022), whether job applicants will stay and perform (Hoffman et al., 2018), and whether released defendants will return to court (Kleinberg et al., 2018). Better information can encourage trust where it will be honoured and discourage it where it will be betrayed. An AI assessment of the counterpart can help to deliver such information effectively. But moving from an assessment to an action may change the recipient's response to that information. A system intended to help people act on a prediction could become less influential precisely when it acts on their behalf. Does giving AI the initiative strengthen reliance on its assessment, or weaken it?

Established forces point in opposite directions. An action that the AI has already initiated resembles a default. People tend to retain defaults through inertia and status-quo bias, and they read them as implicit recommendations (Samuelson and Zeckhauser, 1988; Madrian and Shea, 2001; Johnson and Goldstein, 2003; McKenzie et al., 2006; Jachimowicz et al., 2019). Letting

[1] This form of human oversight, in which the AI acts and a person can confirm or override, is also what regulation requires for high-risk AI systems (European Parliament and Council of the European Union, 2024).

the AI act may also allow people to shift responsibility for an outcome that depends on another person's trustworthiness, a motive that sustains delegation in many settings and that is particularly relevant when trusting puts people at risk of betrayal (Bartling and Fischbacher, 2012; Hamman et al., 2010; Bohnet et al., 2008). Both forces predict that AI initiative should increase reliance on its assessment. Two other forces predict the opposite. First, people value being the authors of their own decisions beyond the payoffs these decisions generate. They forgo earnings to keep decision rights and to retain control, and they derive utility from the procedures by which outcomes come about (Bartling et al., 2014; Fehr et al., 2013; Owens et al., 2014; Frey et al., 2004). Initiating a decision is valued separately from holding the final authority over it (Sloof and von Siemens, 2015), and an AI that presets an action takes this initiative even when the person retains final authority. Second, people resist being steered. Control exerted by another party can crowd out the behaviour it aims to induce (Falk and Kosfeld, 2006), people use algorithms more readily when they keep some control over the outcome (Dietvorst et al., 2018), and they under-delegate decisions even to AI that outperforms them (Ivanova-Stenzel and Tolksdorf, 2025). Both forces predict that AI initiative should reduce reliance on its assessment. Because they attach to confirming an action that the AI has set in motion rather than to the content of the assessment, they predict this reduction whether the assessment encourages trust or warns against it. Which of these forces dominates is an empirical question with direct implications for how AI systems are deployed as decision support systems.

We answer this question in a preregistered experiment with 1,596 UK-based participants recruited through Prolific (Palan and Schitter, 2018).[2] Participants play a binary trust game (Bohnet and Zeckhauser, 2004), described in neutral or explicit trust language, in which a trustor decides whether to transfer an endowment that the trustee can reciprocate after multiplication with factor 2.5. We randomly assign a machine-learning classifier's assessment of whether the trustee will return money, the *AI assessment*, to one of three roles: in *Predict*, it is shown as a prediction of the trustee's behaviour; in *Advise*, as a recommendation; and in *Initiate*, the AI presets the trust decision, which the trustor can confirm or override at no cost. In *Baseline*, the assessment is generated but not shown. Initiate is the oversight arrangement described above, in which a person can override what the AI has set in motion, and it is the mildest form of the AI taking active control: the trustor still makes the same binary choice, but the AI takes the first step. Because the assessment, the available actions, and final authority are identical across AI treatments, differences in reliance identify the effect of the AI's role alone.

Participants rely heavily on the AI. Access to the assessment raises the share of choices that follow it from 48.7% in Baseline to 82.7% in Predict and 83.4% in Advise. When the AI takes the initiative, following falls to 74.5%, a statistically significant 8–9 percentage points below both

[2]The design, hypotheses, and analysis plan were preregistered before data collection in the OSF Registry.

other roles. The most agentic AI implementation in our experiment thereby forfeits about a quarter of its potential influence on behaviour, although participants receive exactly the same information. We call this reduction the *AI agentic penalty*, agentic in the literal sense that the AI initiates the consequential action while the person retains final authority. Prediction and advice produce almost identical reliance, so the preregistered ordering holds at the step to initiative but not at the step from prediction to advice.

The penalty is symmetric. Relative to prediction and advice, Initiate lowers trust by 7.3 percentage points after an assessment that encourages trust and raises it by 8.4 points after one that warns against it. The AI's initiative thus weakens responsiveness in both directions, contrary to status-quo bias and responsibility shifting. The penalty holds in both frames and across specifications. Beliefs and norms account for four fifths of the explained variation across participants. A simple framework separates two channels: a lower weight on the assessment and a procedural cost of ceding the initiative. Beliefs respond less to the assessment when the AI takes the initiative (6.7 against 14–18 percentage points), but the penalty persists conditional on reported beliefs, which fits the procedural cost, as does a stated preference against delegation among participants without an AI role. Perceived accuracy and trait reactance do not explain the pattern.

Our contribution connects two strands of research. The first concerns human–AI interaction: algorithm aversion and appreciation, advice taking, and the allocation of functions between people and machines. It is documented that people can discount algorithms after observing errors, place more weight on algorithmic than human advice, and become more willing to use an algorithm when allowed even limited adjustment (Dietvorst et al., 2015; Logg et al., 2019; Prahl and Van Swol, 2017; Dietvorst et al., 2018). More generally, performance information and the monetary cost of retaining control also shape algorithm use (Ivanova-Stenzel and Tolksdorf, 2024). In financial delegation experiments, however, it has also been shown that reluctance to use algorithms is far from universal (Germann and Merkle, 2023; Holzmeister et al., 2023). In the field, most judges who override an algorithm's bail recommendation perform worse than the algorithm (Angelova et al., 2026). Such overrides can reflect private information or error. In our experiment, participants hold no information beyond the assessment, so differences in overriding reflect the AI's role alone. We identify a further determinant of reliance on AI assistance: reliance changes when an unchanged source moves from supplying information to initiating an action. The human-factors literature makes the related distinction between stages and levels of automation, documenting trade-offs between routine performance, situation awareness, and responses to system failures (Endsley and Kiris, 1995; Onnasch et al., 2014). Our experiment studies the allocation of initiative in an incentivised decision whether to trust another person, and measures its effect on reliance before participants observe any feedback on the AI's performance.

The second strand of literature concerns autonomy and delegation. Decision rights can have

intrinsic value, and people may sacrifice expected earnings to retain control (Frey et al., 2004; Bartling et al., 2014; Owens et al., 2014). Under-delegation to both superior human and AI agents likewise indicates preferences for decision autonomy extending beyond attitudes towards algorithms (Ivanova-Stenzel and Tolksdorf, 2025). Delegation can also change responsibility for the resulting allocation (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011), while organisational research distinguishes initiating and implementing a decision from holding final authority (Sloof and von Siemens, 2015). We show that an economically meaningful behavioural difference arises even when the human retains final authority. Existing work on decision rights varies who holds the authority to decide. Our design holds that authority fixed and varies only who initiates the action. Preserving final authority need not make AI initiative behaviourally equivalent to prediction or advice: people respond to the initiative, not only to the authority. Status-quo bias and responsibility shifting would predict the opposite, which makes the direction of the effect informative about which forces dominate when an AI takes the initiative.

Closest to our design, Özer et al. (2018) compare information sharing, advice provision, and delegation in a customer–supplier relationship. In their setting, the supplier has its own stake in the outcome, and delegation hands the decision to the supplier, so moving across treatments changes incentives and the allocation of authority together with the form of assistance. Our AI has no stake in the transaction, and the trustor retains final authority in every treatment, so only the AI's role varies. Related work studies how unequal access to AI affects perceptions of a human counterpart (Shen et al., 2025), how trustees respond to delegation to humans rather than machines (Hiepler et al., 2026), and how access to ChatGPT changes the promises that trustees make and keep (Greevink et al., 2026). We focus on the trustor and vary whether the same AI predicts, advises, or takes the initiative. This isolates how the delivery of assistance changes the use of information about another person.

The resulting contribution is evidence that the allocation of initiative reduces the use of AI assessments in interpersonal exchange. Prediction quality, available actions, and final authority do not by themselves determine reliance: the role assigned to the predictor also matters. In our setting, an AI that takes the initiative loses about a quarter of the influence that the same AI exerts when it predicts or advises. Our experiment shows that the AI's initiative weakens reliance on the AI irrespective of whether the assessment encourages trust or warns against it. Whether the resulting choices improve welfare depends on the quality of the assessment and the decision-maker's preferences, but the design lesson already follows: the deployment role of an AI system, and not only its accuracy, governs how much its information is used.

Section 2 presents the experiment. Section 3 develops the framework and preregistered hypotheses, and Section 4 sets out the empirical strategy. Section 5 reports the agentic penalty, its decomposition by assessment direction, robustness, heterogeneity, and potential mechanisms. Sec-

tions 6 and 7 discuss the implications and conclude.

2 Experimental Design

The experiment was designed to identify how the institutional role assigned to a fixed AI assessment affects following behaviour. Its design, hypotheses, and analysis plan were preregistered before data collection.[3] In each of the four treatments, Participant A ultimately chose between the same two consequential actions in the same binary trust game. A machine-learning classifier assessed whether Participant B would return money. What changed across treatments was whether this classifier assessment was unavailable, presented as a prediction, translated into advice, or implemented as an action that the participant could retain or override. The neutral and natural trust frames provided an independently designed wording variation for robustness. The complete experimental instructions can be found in Appendix D.

2.1 Trust Game

We used the binary trust game of Bohnet and Zeckhauser (2004). Participant A received 12 experimental points (EP) and decided whether to keep them or send them to Participant B. If A kept the endowment, the payoffs were 12 EP for A and 0 EP for B. If A sent it, the amount was multiplied by 2.5 and B received 30 EP. Participant B then either returned 14 EP, yielding payoffs of 14 EP for A and 16 EP for B, or kept all 30 EP, yielding 0 EP for A and 30 EP for B. Figure 1 displays the game.

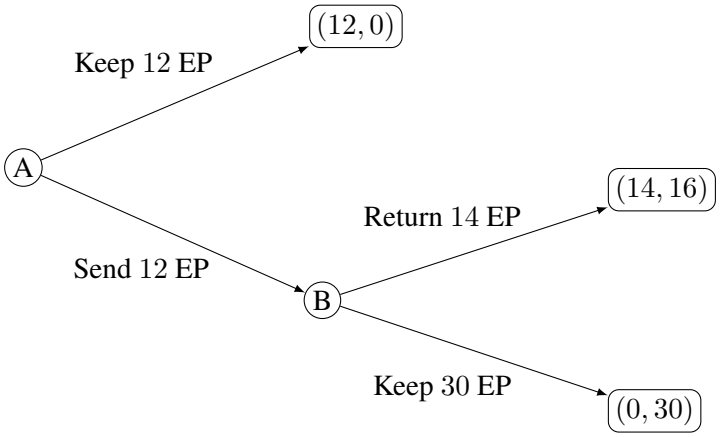


Figure 1: Binary trust game

Note: Terminal payoffs are shown in experimental points, ordered as Participant A, Participant B.

[3]The preregistration is available in the OSF Registry.

If Participant A assigns probability q to B returning 14 EP, trusting maximises expected monetary payoff when $14q \geq 12$, or $q \geq 6/7$. Trusting therefore pays only if Participant A is quite confident that this particular Participant B will return. An assessment of whether Participant B will return speaks directly to that probability, which makes it relevant to Participant A's decision.

Participant B's decision is binary: return 14 EP or keep all 30 EP. The AI's information could therefore be summarised as a single yes-or-no assessment of whether Participant B would return, which is what a standard binary classifier delivers. The same assessment could then be presented as a prediction, a recommendation, or an action without changing its content. At the same time, interpersonal trust provided a consequential context with uncertainty about the other participant's motives and future actions in which AI could assist Participant A by predicting the other person's behaviour, recommending an action, or acting on Participant A's behalf.

We used the strategy method (Selten, 1967). Every participant first chose as Participant A and then chose as Participant B conditional on A having sent the endowment. Every participant thus made a treated decision as Participant A, which is our object of interest, and also left a recorded Participant B decision that could be paired with any partner for payment. After both choices, one member of each matched pair was randomly assigned the A role and the other the B role for payment.

2.2 AI Assessment

For each matched partner, a machine-learning classifier generated a binary assessment of whether that partner would return 14 EP in the role of Participant B. Participants were introduced to it as an AI tool. Below, *AI tool* refers to the tool as presented to participants and *classifier* to the underlying model. Every assessment shown to participants came from this classifier. The assessment itself was invariant to treatment. Treatment changed only how it entered Participant A's decision environment.

The classifier was trained on roughly 16,000 return decisions from ten previous trust-game studies and from our own pilot study described below, all listed in Appendix Table A.1. It is a CatBoost classifier that used fixed game parameters and the counterpart's pre-treatment age, gender, student status, continent, economics background, education, and ethnicity indicator. Its accuracy on an independent hold-out sample of the training data was 89.86%. This accuracy was disclosed to Participant A before the decision in the three AI treatments and, in the post-choice questionnaire, to Baseline participants as well. Participants were told that the tool had been trained on decisions taken by participants in earlier experiments of this kind, which was the case.

Throughout the remainder of the paper, *assessment* denotes this binary judgement of the classifier. Only Participant A saw it, and only in the three AI treatments. Depending on the treatment,

Participant A saw a positive assessment as a prediction that Participant B would return 14 EP, as a recommendation to send 12 EP, or as an initial decision to send 12 EP. A negative assessment correspondingly appeared as a prediction that Participant B would not return 14 EP, a recommendation not to send 12 EP, or an initial decision not to send 12 EP. Thus, the informational content was held fixed while its institutional form varied across treatments.

In the experimental sample, the classifier had little out-of-sample discriminatory power. Counterparts returned 14 EP in 69.7% of cases after a positive assessment and in 66.6% after a negative assessment (Appendix B.12). Participants could not learn this before deciding. They did not observe the counterpart's characteristics and received no feedback until after the questionnaire. The assessment, its disclosed accuracy, and its relation to counterpart behaviour were identical across treatments. Participants who departed from the assessment therefore cannot have done so because they detected classifier errors, and the treatment comparisons isolate how the AI's role changes the weight placed on information that participants believed to be highly accurate.

2.3 AI Role Treatments and Language Frame

Participants were assigned to one of four AI-role treatments. Figure 2 summarises the treatment comparisons.

Treatment	Classifier assessment	AI's institutional role	Consequential choice set
Baseline	Generated but hidden	No AI information shown	Trust or do not trust
Predict	Shown to participant	Predicts whether Participant B will return	Trust or do not trust
Advise	Shown to participant	Advises whether Participant A should trust	Trust or do not trust
Initiate	Shown through the initial action	Initiates the trust decision; Participant A confirms or overrides	Trust or do not trust

Figure 2: Experimental variation in the AI's institutional role

In all treatments, including the *Baseline*, all participants were told that the AI tool existed before making their decision as Participant A. Participants in the Baseline, however, had no access to it, i.e., the classifier assessment was generated and recorded but remained hidden. In the *Predict* treatment, the assessment was shown as a prediction of whether Participant B would return 14 EP. In the *Advise* treatment, the same assessment was translated into a recommendation to send or not send. In the *Initiate* treatment,[4] it was translated into an initial action that would send

[4]The preregistration labels this treatment *Perform*. We use *Initiate* throughout; the treatment is otherwise as preregistered.

or not send. In this sense, the AI preset the decision: participants saw the AI tool's decision and confirmed it ("Stick with the AI tool's decision") or overrode it ("Override the AI tool's decision"), each option labelled with the corresponding consequential action. The Initiate treatment therefore required active confirmation rather than relying on a passive default. Read as a recommendation with a default, Initiate should raise following: defaults, the implicit recommendation they carry, and responsibility shifting all push in that direction. We find the opposite. The Initiate treatment gave the AI an agentic role, but did not remove participants' final control or change the set of feasible consequential actions. It is the mildest form of AI initiative. The AI takes the first step, so a participant who confirms it no longer authors the choice alone, but every participant sees its assessment and makes the same binary choice as in Predict and Advise. This mildness is what keeps the comparison with prediction and advice clean. A stronger form of initiative, delegating the entire decision to the AI before seeing its choice, was implemented in our pilot study and points in the same direction. In every treatment, Participant A made one choice on the same decision screen, so the AI's role did not save participants any effort. The instructions described the three AI treatments in parallel wording, with two differences. The AI tool was introduced as programmed to "inform", "advise", or "support" the participant, and its analysis was described as enabling "well-informed" predictions and recommendations in Predict and Advise but "informed" decisions in Initiate. The disclosed accuracy of 89.86%, its explanation, and the comprehension questions about it were identical across the three AI treatments (Appendix D), and perceived classifier accuracy does not differ between them.

Reliance on algorithms can depend on whether a task is perceived as objective or subjective (Castelo et al., 2019; Chugunova and Sele, 2022). We therefore crossed the four AI roles with two language frames to test whether the role effects depended on casting the same actions in neutral transfer terms or explicit trust terms. The neutral frame referred to sending and returning points. The natural trust frame described the corresponding choices as trusting, distrusting, and being trustworthy. The AI tool's messages used the same frame as the game. Both members of a pair were assigned the same treatment, and each participant made the Participant B decision after their own decision as Participant A. Participants deciding as B had therefore experienced the same AI role as their counterpart, but they were not told which assessment their counterpart had received.

2.4 Assignment, Procedure, and Sample

Treatment and frame were adaptively randomised at the pair level among the eight experimental cells, using inverse cell counts to maintain balance as participants arrived. The pair identifier is retained for clustered robustness inference. Because every participant made the Participant B

decision using the strategy method, incomplete partners did not invalidate recorded choices.[5]

The experiment was programmed in oTree (Chen et al., 2016) and conducted online with UK-based participants recruited through Prolific (Palan and Schitter, 2018). Participants who failed both embedded attention checks were stopped by the software, and non-completers were excluded. The analysis sample contains 1,596 participants. Comprehension questions instead supplied corrective feedback and required a correct response before a participant could continue. These questions were not an exclusion criterion. The median participant was 41 years old. Mean earnings were £3.34, comprising a £1.50 participation fee and a performance-based bonus at eight EP per pound.

After the trust-game choices, participants completed a questionnaire. It measured beliefs about how many other Participants A sent and how many other Participants B returned, social-appropriateness judgements inspired by Krupka and Weber (2013), perceived AI accuracy, willingness to use different AI roles, attitudes towards AI (Sindermann et al., 2021), trait reactance, a general disposition to resist perceived restrictions on one's freedom of choice (Hong and Faedda, 1996), and standard demographic and behavioural-economic measures.[6] We distinguish these post-treatment measures from covariates observed before assignment in the empirical analysis.

2.5 Pilot Study

Before the main study, we ran a pilot study on Prolific in April 2024. It served to calibrate payoffs, timing, and the power calculation, and to collect return decisions from the participant pool of the main study on which to train the classifier. The second purpose made the pilot larger than calibration alone would have required: 253 participants completed the four AI-role treatments. The pilot used a different payoff calibration, an earlier version of the classifier, and assignment of treatment and frame by session (Appendix C).

Most importantly, the pilot implemented the most far-reaching form of AI initiative, ex-ante delegation. Participants decided whether to delegate the decision to the AI tool before seeing what the tool would do. If they delegated, they saw the tool's decision, which was implemented without the option to override. If they did not delegate, they decided themselves without seeing the assessment. The outcome of this design is the delegation choice. It cannot compare reliance on the same information across roles, because whether a participant receives the assessment depends on

[5]For payoff settlement, incomplete cases were rematched where an observed Participant B choice preserved the truth of the previously displayed classifier assessment. Remaining unmatched participants were informed that their original partner had not completed, and a partner decision was drawn for payment. These procedures occurred after Participant A's treatment, assessment, and choice had been recorded.

[6]The latter include trusting disposition (Gefen and Straub, 2004; Berger et al., 2021), personal innovativeness in information technology (Agarwal and Prasad, 1998), locus of control (Kovaleva, 2012), risk tolerance (Dohmen et al., 2011), belief in free will (Harms et al., 2017), and altruism (Falk et al., 2018). Appendix D.9 contains the questions and response scales.

that choice. The main study therefore lets the AI initiate the action while every participant sees it and retains final authority. The pilot nonetheless points in the same direction as the main results.

3 Conceptual Framework

The design varies whether an AI assessment informs, recommends, or initiates a consequential action. This section provides a deliberately minimal framework linking that role variation to the two outcomes used in the experiment and the analyses: following the AI and choosing to trust.

Let $a \in \{-1, +1\}$ denote the classifier assessment, where $+1$ indicates that the prospective trustee is predicted to return and -1 indicates that the trustee is predicted not to return. After observing the assessment in role $r \in \{P, A, I\}$, where P, A, and I denote Predict, Advise, and Initiate, participant i forms the latent belief index

$$q^*_{ir}(a) = \mu_i + \lambda_r \delta a, \tag{1}$$

where μ_i is the participant's prior, $\delta > 0$ is the perceived informational content of the assessment, and $\lambda_r \geq 0$ is the weight the participant places on it in role r. Only the product $\lambda_r \delta$ is identified. Because the assessment and its disclosed accuracy are identical across roles, δ carries no role index, and differences between roles operate through λ_r. Perceived accuracy, which participants reported after their choice, offers an indirect check on this restriction. The perceived return probability is the latent belief index censored at the bounds of the unit interval,

$$q_{ir}(a) = \min\big\{\max\{q^*_{ir}(a), 0\},\, 1\big\},$$

so that $q_{ir}(a) = q^*_{ir}$ when $q^*_{ir}(a) \in [0, 1]$.

The expected monetary-payoff difference between trusting and not trusting is $14 q_{ir}(a) - 12$. Participants may also care about who authors the action that is carried out. Let $\psi \geq 0$ denote the value participants attach to being the author of their own decisions (Bartling et al., 2014; Frey et al., 2004), and let $\phi_r \in [0, 1]$ denote the share of authorship of a choice in line with the assessment that passes to the AI in role r. A choice against the assessment always leaves the participant as its sole author. Under prediction, the participant turns the AI's information into a choice of their own, so $\phi_P = 0$. Under Initiate, the AI has set the action and the participant confirms it, so ϕ_I is large. Following the AI may also bring benefits $b_r \geq 0$. Status-quo bias and the implicit recommendation of a default require an action that the AI has set in motion, whereas responsibility can also shift to an AI whose recommendation one follows. Hence $b_P = 0$, $b_A \geq 0$ is small, and $b_I \geq b_A$. Following the assessment therefore carries the net procedural term $c_r = \psi \phi_r - b_r$. Because following means

trusting after $a = +1$ and not trusting after $a = -1$, the latent utility difference between trusting and not trusting is $14q_{ir}(a) - 12 - c_r a + \varepsilon_i$. For participants whose belief index lies in the unit interval, this becomes

$$D_{ir}(a) = 14\mu_i - 12 + (14\lambda_r\delta - c_r)\, a + \varepsilon_i. \tag{2}$$

At the bounds, the assessment moves the belief no further, which can only weaken the response of the trust choice to the assessment. Because $q_{ir}(+1)$ is non-decreasing and $q_{ir}(-1)$ non-increasing in λ_r, the comparative statics below hold for all participants and are strict for those with interior beliefs. Under prediction, $c_P = 0$, and under advice, $c_A = \psi\phi_A - b_A$ is small if a recommendation leaves the choice with the participant and offers little room for shifting responsibility. Under Initiate, the sign of $c_I = \psi\phi_I - b_I$ is not restricted a priori, because competing forces bear on it. Status-quo bias, the implicit recommendation carried by a default, and responsibility shifting for the outcome raise b_I and make confirmation attractive (Samuelson and Zeckhauser, 1988; McKenzie et al., 2006; Bartling and Fischbacher, 2012). The value of authorship makes confirmation costly through $\psi\phi_I$ (Bartling et al., 2014; Owens et al., 2014; Sloof and von Siemens, 2015). Define $\omega_r = 14\lambda_r\delta - c_r$. A smaller ω_r means that the final trust choice responds less to the direction of the same assessment. This can arise because the AI's initiative lowers the weight on the assessment, because the authorship cost of confirming outweighs its benefits, or through both channels. If status-quo bias and responsibility shifting dominated instead, the AI's initiative would raise ω_I above its values under prediction and advice and increase following. The penalty we measure is therefore a net effect. Because $b_I \geq 0$, the authorship cost $\psi\phi_I$ is at least as large as the net term c_I, so the aversion to the AI's initiative is at least as large as its measured net effect.

Naturally, the framework collapses to the standard prediction for selfish and rational players in the Baseline. If it is common knowledge that players are selfish, Participant B never returns, and Participant A holds the belief $\mu_i = 0$ with certainty. No assessment can move a belief held with certainty, so $\lambda_r = 0$. Without a value of authorship or benefits of following, $\psi = 0$ and $b_r = 0$, so that $c_r = 0$, and without idiosyncratic components, $\varepsilon_i = 0$. Then $D_{ir}(a) = -12$ in every role, and Participant A never trusts. The AI's role remains irrelevant when Participant A is uncertain about Participant B's type. A rational, selfish Participant A then updates on the assessment, but in the same way in every role, because the information is identical, so that $\lambda_r = \lambda$. With $c_r = 0$, $\omega_P = \omega_A = \omega_I$. An agentic penalty therefore requires a departure from this benchmark: a weight on the assessment that depends on the AI's role, a procedural term, or both.

The empirical implications do not depend on a particular distribution for ε_i. Let

$$p_r^+ = \Pr(\text{Trust} = 1 \mid r, a = +1), \qquad p_r^- = \Pr(\text{Trust} = 1 \mid r, a = -1),$$

and define role-specific responsiveness as $\rho_r = p_r^+ - p_r^-$. The assessment-standardised probab-

ility of following the AI within role r gives positive and negative assessments equal weight by construction:

$$\widetilde{C}_r = \frac{1}{2}\left[p_r^+ + (1 - p_r^-)\right] = \frac{1}{2}(1 + \rho_r), \tag{3}$$

Weighting both assessments equally is a normalisation, not an assumption. It makes following independent of how often each assessment occurs, so that $\widetilde{C}_r$ depends only on responsiveness. The preregistered Follow rate instead weights the two assessments by their realised frequencies. Because these frequencies are balanced across treatments, treatment differences in Follow measure treatment differences in responsiveness.

Let C_r denote the realised Follow rate. The comparisons between the three AI roles are the pairwise differences

$$\Delta_{PA} = C_P - C_A, \qquad \Delta_{PI} = C_P - C_I, \qquad \Delta_{AI} = C_A - C_I. \tag{4}$$

A positive value means that reliance on the same assessment falls as the AI's role extends further into the decision.

3.1 Preregistered Hypotheses

The preregistered hypotheses side with the decision-rights literature. If people value being the author of their decisions ($\psi > 0$), a larger role for the AI passes more of that authorship to the AI ($\phi_P < \phi_A < \phi_I$), and the benefits of confirmation do not outweigh this cost, the assessment has less net influence as the AI's role moves from prediction to recommendation to initiative:

$$\omega_P > \omega_A > \omega_I.$$

The framework does not impose this ordering. If status-quo bias or responsibility shifting dominated, the ordering would reverse at the step to initiative. The hypotheses are therefore a directional prediction that the experiment tests against this alternative. Because priors and the perceived informational content of the assessment are distributed identically across randomly assigned roles, equation (2) implies $\widetilde{C}_P > \widetilde{C}_A > \widetilde{C}_I$ under this ordering. The empirical counterparts are the preregistered Follow rates C_P, C_A, and C_I. This yields the two preregistered hypotheses:

Hypothesis 1. *Participants in Predict are more likely to follow the AI assessment than participants in Advise and participants in Initiate.*

Hypothesis 2. *Participants in Advise are more likely to follow the AI assessment than participants in Initiate.*

Together, Hypotheses 1 and 2 impose the strict ordering $C_P > C_A > C_I$, that is, $\Delta_{PA} > 0$, $\Delta_{PI} > 0$, and $\Delta_{AI} > 0$. In the framework, $\Delta_{PA} > 0$ requires a recommendation to pass some authorship to the AI ($\phi_A > \phi_P$), whereas $\Delta_{PI} > 0$ and $\Delta_{AI} > 0$ require a preset to pass more authorship to the AI ($\phi_I > \phi_A$).

3.2 Decomposition by Assessment Direction

A difference in responsiveness can be examined separately for the two assessment directions. Write $\bar{p}_r = (p_r^+ + p_r^-)/2$. Relative to a comparison role r, the shifts away from the assessment under Initiate, a reduction in trust after a positive assessment and an increase in trust after a negative assessment, are

$$d_r^+ = p_r^+ - p_I^+ = (\bar{p}_r - \bar{p}_I) + \frac{1}{2}(\rho_r - \rho_I), \tag{5}$$

$$d_r^- = p_I^- - p_r^- = -(\bar{p}_r - \bar{p}_I) + \frac{1}{2}(\rho_r - \rho_I). \tag{6}$$

Consequently, $d_r^+ = d_r^-$ when the AI's initiative changes responsiveness while leaving the midpoint of trust probabilities unchanged. A symmetric reduction in responsiveness is thus the observable implication of a role-induced reduction in how strongly choices move with the AI assessment, rather than a general shift towards either more or less trust.

The decomposition also separates the competing forces introduced above. A uniform reduction in ω_I, whether through λ_I or through $c_I > 0$, lowers responsiveness in both directions and leaves the midpoint unchanged, so that $d_r^+ = d_r^- > 0$. Status-quo bias raises adherence to the AI-initiated action in both directions and implies $d_r^+ = d_r^- < 0$. Responsibility shifting operates mainly when the AI's action exposes the participant to betrayal, that is, after a positive assessment. It raises p_I^+ without a matching change in p_I^-, shifts the midpoint, and implies $d_r^+ < d_r^-$. The sign and symmetry of the two shifts away from the assessment therefore discriminate between a direction-neutral reduction in the weight on the assessment and these directional alternatives.

4 Empirical Strategy

Our empirical strategy separates the preregistered treatment comparison from the decomposition of the effect by assessment direction and from subsequent explanatory analyses.

4.1 Preregistered Outcome and Tests

For the three AI roles, *Follow* equals one when Participant A sends after a positive classifier assessment or does not send after a negative assessment. In Baseline, the analogous measure records alignment with an assessment that was generated but not shown to participants. It therefore does not measure following. Our confirmatory tests compare Follow rates using two-sided Boschloo tests. Our implementation uses the ordinary two-sided Fisher exact p-value as the ordering statistic.

The preregistered power calculation defines a five-test Bonferroni family: Predict–Baseline, Advise–Baseline, Initiate–Baseline, Predict–Advise, and Advise–Initiate. We therefore use a per-test threshold of $0.05/5 = 0.01$. Because Hypothesis 1 also explicitly compares Predict with Initiate, we report that treatment comparison alongside the five preregistered pairs even though it is not among them. Appendix B.1 maps the full family and this additional hypothesis-implied treatment comparison.

4.2 Estimands

We first measure how much the assessment moves behaviour at all. Baseline alignment C_B records how often choices would have matched the assessment without access to it. The excess $C_r - C_B$ therefore measures the AI's influence on behaviour in role r. Its three values are the Baseline treatment comparisons of the preregistered family.

Hypotheses 1 and 2 concern the three pairwise differences between the AI roles,

$$\Delta_{PA} = C_P - C_A, \qquad \Delta_{PI} = C_P - C_I, \qquad \Delta_{AI} = C_A - C_I,$$

which are the primary empirical measures. We estimate all three rather than pool any two roles.

Two descriptive summaries complement the primary differences. First, the share of the AI's influence that the initiative forfeits relative to prediction and advice is

$$1 - \frac{C_I - C_B}{\frac{1}{2}(C_P + C_A) - C_B}.$$

Second, because the preregistered ordering implies two successive steps, we test whether the advice step $C_P - C_A$ and the initiative step $C_A - C_I$ are of equal size. A rejection with a larger initiative step locates the penalty at the point where the AI takes the initiative. Both summaries are exploratory and use a linear probability model with robust standard errors.

To reveal what the Follow indicator compresses, we estimate trust probabilities separately by

AI role and classifier assessment. For each AI role r, the assessment difference

$$\rho_r = \Pr(\text{Trust} = 1 \mid r, a = +1) - \Pr(\text{Trust} = 1 \mid r, a = -1)$$

measures how strongly the trust choice varies with the classifier assessment. Random assignment of AI role, together with the balanced composition of assessments across treatments, identifies the treatment comparisons between these differences. The classifier assessment itself was generated from counterpart characteristics rather than independently randomised, so within-treatment differences describe responsiveness to classifier-generated assessments.

We assess symmetry by comparing Initiate with a comparison role r. The shift away from a positive assessment is

$$d_r^+ = p_r^+ - p_I^+,$$

and the shift away from a negative assessment is

$$d_r^- = p_I^- - p_r^-.$$

The Wald test of $d_r^+ = d_r^-$ asks whether the AI's initiative reduces responsiveness equally in the pro-trust and anti-trust directions. The appropriate comparison role depends on how Predict and Advise compare, so this test is exploratory. It does not replace the preregistered treatment comparisons.

4.3 Specifications and Inference

Binary outcomes are coded zero and one throughout the regressions. Our adjusted models use probit specifications and report average predicted probabilities, average marginal effects, and treatment differences in probability units. The role-by-assessment model takes the form

$$\Pr(Y_i = 1) = \Phi(\alpha + \boldsymbol{R}_i'\boldsymbol{\beta} + \gamma a_i + \boldsymbol{R}_i' a_i \boldsymbol{\kappa} + \eta F_i + \boldsymbol{X}_i'\boldsymbol{\theta}), \tag{7}$$

where $\boldsymbol{R}_i$ contains AI-role indicators, F_i is the randomised language frame, and $\boldsymbol{X}_i$ is the relevant covariate set. Baseline is retained as a benchmark in which the assessment was generated but not shown. We use robust standard errors in the main specifications and repeat the pre-treatment model with errors clustered by pair-level assignment group.

The sequence of specifications follows covariate timing. The unadjusted or minimally adjusted model establishes the design-based result. The preferred adjusted benchmark includes treatment assignments, frame, and covariates observed before treatment. We then add post-treatment demographics, technology attitudes, beliefs and norms, and general attitudes in sequence. The final

model is the full specification, but it is not used to replace the pre-treatment benchmark. We explicitly discuss any material change in sign, magnitude, or statistical inference. Otherwise, we describe estimates as stable after sequential inclusion of covariates. The full specification also serves one interpretive purpose. Because it conditions on reported beliefs and social-appropriateness judgements, it indicates whether the agentic penalty is absorbed by what participants report believing. It is a descriptive comparison, not a mediation decomposition.

Framing, zero-mismatch comprehension, and pair-clustered estimates are robustness analyses. The grouped Shapley–Owen decomposition is descriptive: it allocates the incremental ordinary R^2 of a linear probability model across prespecified covariate blocks. Post-choice beliefs, perceived accuracy, role preferences, and reactance are used to distinguish potential explanations, not as separately randomised mediators. Unless explicitly tied to a preregistered hypothesis above, these analyses are secondary or exploratory.

5 Results

We first establish how strongly participants rely on the AI assessment and the preregistered effect of the AI's role on following. We then turn to the underlying trust choices and test whether the effect is symmetric with respect to the AI's assessment about Participant B's returning decision. Subsequently, we establish robustness, describe observable heterogeneity, and assess potential mechanisms.

5.1 The AI Agentic Penalty

The analysis sample contains 1,596 participants. Our preregistered outcome, *Follow*, records whether the participant's final action matches the classifier assessment. Table 1 reports rates by treatment and the three treatment differences implied by the hypotheses.

Participants rely heavily on the AI assessment. In Baseline, where the assessment was generated but not shown, participants' choices matched it in 48.7% of cases. Access to the assessment raised this alignment by 34.0 percentage points in Predict, 34.6 points in Advise, and 25.8 points in Initiate. All three differences belong to the preregistered family and are statistically significant ($p < 0.001$, Boschloo tests; Appendix Table B.1). We refer to this gain over Baseline as the AI's influence on behaviour. The hypotheses concern how the AI's role changes it.

Follow is 82.7% in Predict, 83.4% in Advise, and 74.5% in Initiate. Predict and Advise are virtually identical ($\Delta = -0.7$ percentage points, $p = 0.849$, Boschloo test). Relative to Initiate, Follow is 8.2 percentage points higher in Predict ($p = 0.005$, Boschloo test) and 8.9 points higher in Advise ($p = 0.002$, Boschloo test). Both meet the preregistered Bonferroni-adjusted threshold

of $\alpha = 0.01$.

Treatment	Follow / alignment (%)	N
Baseline (assessment not shown)	48.7	398
Predict	82.7	399
Advise	83.4	403
Initiate	74.5	396
Preregistered treatment comparison	**Difference (pp)**	**Boschloo p**
Predict–Advise (H1)	-0.7	0.849
Predict–Initiate (H1)	8.2	0.005
Advise–Initiate (H2)	8.9	0.002

Note: Follow equals one when the participant's final action matches the classifier assessment. In Baseline, the assessment was generated but not shown to participants, so the reported quantity is alignment with the AI assessment. The preregistered multiplicity family and the two-sided Boschloo test are described in Appendix B.1.

Table 1: Preregistered effect of AI role on following

Hypothesis 1 predicted more following in Predict than in both Advise and Initiate. It is partially supported: Predict exceeds Initiate but not Advise. Hypothesis 2 predicted more following in Advise than in Initiate and is supported. The evidence therefore locates the penalty at the move from prediction or advice to AI initiative, not at the move from prediction to advice. The two steps of the preregistered ordering also differ in size. The initiative step, Advise minus Initiate, exceeds the advice step, Predict minus Advise, by 9.5 percentage points ($p = 0.043$, linear probability model with robust standard errors). Reliance does not decline gradually as the AI takes over more of the decision. It falls once the AI takes the initiative.

We call this reduction the *AI agentic penalty*: participants rely less on the same AI assessment when the system initiates the action on their behalf than when it predicts another participant's behaviour or recommends their own action. The system's information and the participant's two consequential actions are held fixed. Only the AI's institutional role varies across treatments. Measured against the AI's influence under prediction and advice, the initiative forfeits 24.9% of it, about one quarter.

Result 1. *Changing the AI's assistance from a prediction or a recommendation to an overridable AI-initiated action reduces following by 8–9 percentage points, about one quarter of the AI's influence on behaviour. This is the AI agentic penalty.*

The Follow measure establishes the phenomenon, but combines trusting after a positive assessment with withholding trust after a negative assessment. We next examine whether the agentic penalty is concentrated in one of these two assessment directions or reflects a general reduction in responsiveness.

5.2 The Penalty by Assessment Direction: A Symmetric Reduction in Responsiveness

Figure 3 reports pre-treatment-adjusted trust responsiveness to the classifier assessment. Panel A shows the positive-minus-negative assessment difference within each treatment. Panel B reports the treatment comparisons between these differences. Responsiveness is substantial in all three AI treatments, but markedly weaker in Initiate than in Predict or Advise. In Baseline, trust barely varies with the assessment, which participants did not observe.

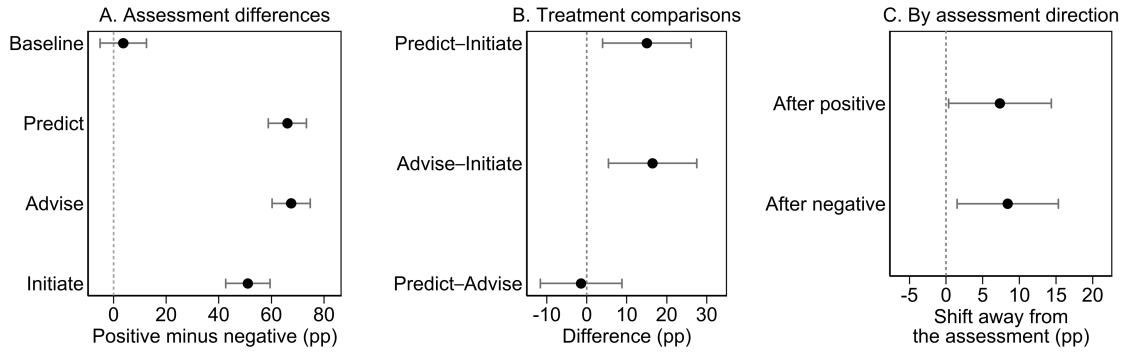


Figure 3: Trust responsiveness to the classifier assessment by AI role and assessment direction

Note: Panel A reports average marginal effects of a positive rather than negative assessment on trust. Panel B reports treatment comparisons between these assessment differences. Panel C reports the shifts away from a positive and a negative assessment under Initiate relative to the equal-weighted average of Predict and Advise. Bars show 95% confidence intervals using robust standard errors. The Baseline assessment was generated but not shown.

Column (1) of Table 2 quantifies these assessment differences. A positive rather than negative assessment raises trust by 66.1 percentage points in Predict and 67.5 points in Advise, but by 51.0 points in Initiate. The Predict–Initiate treatment comparison is 15.0 percentage points ($p = 0.008$), and the Advise–Initiate treatment comparison is 16.5 points ($p = 0.004$). Predict and Advise are again statistically indistinguishable ($p = 0.786$). Responsiveness under Initiate is thus 23.6% lower than under prediction and advice, about one quarter, matching the share of influence forfeited in the Follow measure.

Because Predict and Advise produce almost identical responsiveness, we compare Initiate with their equal-weighted average, so that $d^{+} = \frac{1}{2}(d_P^{+} + d_A^{+})$ and $d^{-} = \frac{1}{2}(d_P^{-} + d_A^{-})$. This pooling follows from the observed similarity of the two roles, not from a grouping fixed in advance. The reduction in responsiveness is symmetric. Relative to this benchmark (Figure 3, Panel C), the Initiate treatment lowers trust after a positive assessment by 7.3 percentage points and raises trust after a negative assessment by 8.4 points. The two shifts away from the assessment differ by

only -1.1 points, and the test of equal magnitudes does not reject symmetry ($p = 0.831$). The AI's agentic role therefore makes participants symmetrically less responsive to its assessment, rather than shifting trust in one direction. This pattern distinguishes between the competing forces. Had status-quo bias dominated, trust would have followed the AI-initiated action more closely in both directions. Had responsibility shifting dominated, participants would have trusted more readily when the AI extended trust on their behalf, producing a larger shift after positive than after negative assessments. Neither pattern appears. The AI's initiative lowers the weight placed on the assessment in both directions.

	(1) Pre-treatment	(2) Full	(3) Pair-clustered
Baseline	0.037	0.015	0.037
	(0.045)	(0.035)	(0.044)
Predict	0.661***	0.444***	0.661***
	(0.037)	(0.042)	(0.037)
Advise	0.675***	0.475***	0.675***
	(0.037)	(0.042)	(0.036)
Initiate	0.510***	0.309***	0.510***
	(0.043)	(0.043)	(0.042)
p(Predict = Advise)	0.786	0.589	0.784
p(Predict = Initiate)	0.008	0.022	0.007
p(Advise = Initiate)	0.004	0.005	0.003
p(symmetry)	0.831	0.397	0.834
Observations	1,596	1,596	1,596

Note: Entries are average marginal effects of a positive rather than negative classifier assessment on the probability of trust; standard errors are in parentheses. Column (1) includes framing and pre-treatment covariates. Column (2) adds the complete sequence of pre- and post-treatment covariates. Column (3) repeats column (1) with errors clustered by pair-level assignment group. The symmetry test compares the shifts away from a positive and a negative assessment between Initiate and the equal-weighted average of Predict and Advise. In Baseline, the assessment was generated but not shown to participants. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2: Trust responsiveness: estimates and treatment comparisons

Following and trust responsiveness therefore show the same phenomenon. The preregistered outcome shows less following in Initiate. The disaggregated choices show a symmetric reduction in responsiveness to assessments that encourage and discourage trust.

Result 2. *The agentic penalty is symmetric across assessment direction: the AI's initiative reduces responsiveness to assessments that encourage trust and to assessments that discourage it by similar magnitudes.*

5.3 Robustness and Scope

The agentic penalty is stable after sequential inclusion of covariates. Figure 4 traces the three focal Follow treatment comparisons across six specifications. Predict and Advise remain alike, while the Predict–Initiate and Advise–Initiate differences remain positive and statistically distinguishable from zero. The preferred benchmark includes only pre-treatment covariates. Later columns add post-treatment measures for completeness. They do not redefine the treatment effect.

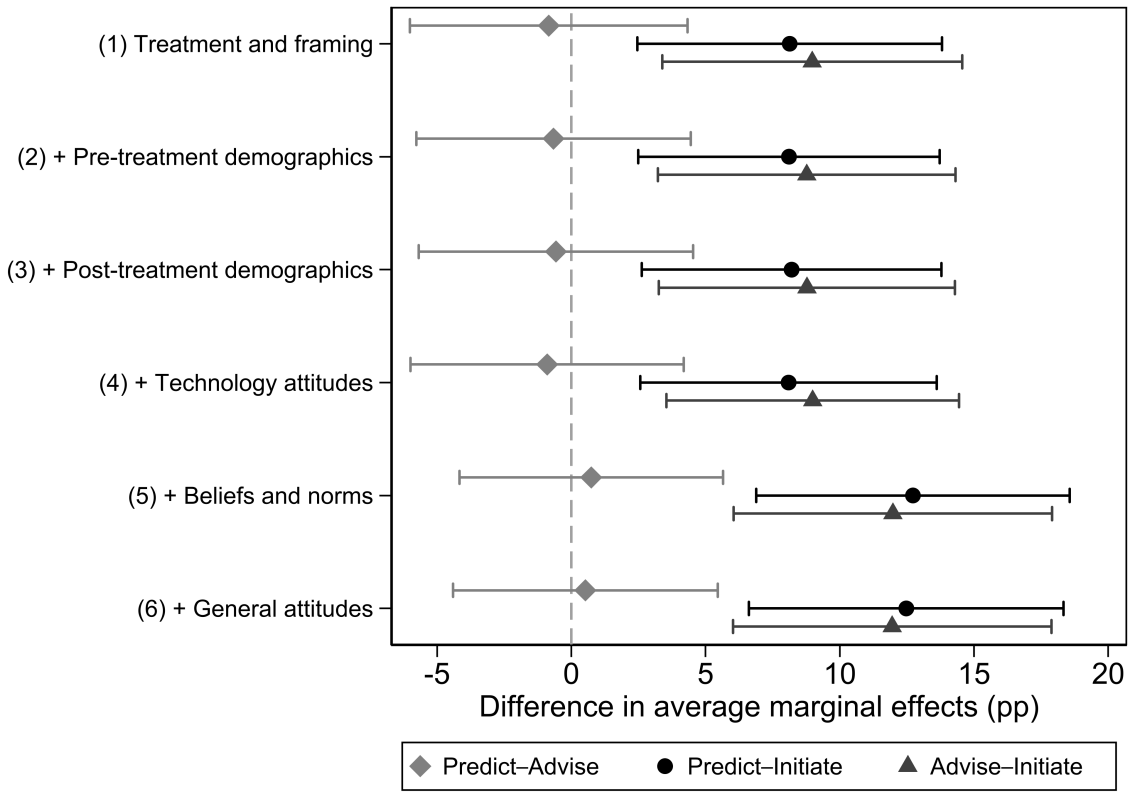


Figure 4: Follow differences under sequential inclusion of covariates

Note: Points are pairwise differences between probit average marginal effects; bars show 95% confidence intervals using robust standard errors. Column (2) is the preferred adjusted specification. Later specifications sequentially add post-treatment covariates and are descriptive.

The trust-responsiveness result is equally stable. The full specification produces positive-minus-negative assessment differences of 44.4, 47.5, and 30.9 percentage points in Predict, Advise, and Initiate. Both Predict–Initiate and Advise–Initiate differences remain positive ($p = 0.022$

and $p = 0.005$), and the pair-clustered pre-treatment estimates leave the conclusion unchanged. The natural trust frame raises following by 5.2 percentage points ($p = 0.020$), but the ordering Predict $\approx$ Advise $>$ Initiate appears in both the neutral and natural trust frames. The penalty is not confined to a particular description of the game (Appendix B.7).

Comprehension questions required a correct answer before participants could continue and were not used to exclude anyone. A deliberately stringent exploratory analysis using only the 584 participants who recorded no mismatch on any attempt preserves the qualitative treatment ordering (Appendix B.8). Together, these checks show that the agentic penalty is not an artefact of covariate selection, pair-level inference, wording frame, or the main comprehension rule.

Our pilot study points in the same direction (Appendix C). It was not preregistered, is smaller, and implemented ex-ante delegation instead of Initiate, so its outcome is the choice whether to delegate rather than Follow. When the AI's role was to take over the decision entirely, 52.5% of participants declined and decided themselves, although this meant deciding without the assessment of a tool presented as 83% accurate. Whether participants delegated did not depend on the assessment, which they had not yet seen. In the main study, retaining the initiative costs participants no information, yet reliance on the assessment still falls when the AI takes the initiative. In both designs, a substantial share of participants resist handing the initiative to the AI.

5.4 Observable Heterogeneity

The preceding estimates establish that covariate adjustment does not explain away the experimental differences. We now ask which participants follow the AI more. Differences in beliefs about other participants' behaviour and in judgements of what is socially appropriate account for most of the variation in Follow that observed characteristics can explain. Demographic background, attitudes towards technology, and general behavioural traits add comparatively little. To quantify this, Figure 5 divides the additional explained variation, that is, the increase in the ordinary R^2 of a linear probability model when the five covariate blocks are added to treatment and framing, among those blocks using an exact grouped Shapley–Owen decomposition. The linear probability model is used here so that the decomposition retains its ordinary variance interpretation.

Treatment and framing alone yield an R^2 of 0.016 among the 1,198 participants in the three AI treatments, as expected for a binary outcome whose treatment differences are a few percentage points. Adding the five covariate blocks raises the R^2 to 0.222, and the decomposition allocates this increment of 0.206. Beliefs and norms account for 79.2% of the increment. Of the additional predictive power supplied by all measured covariates, roughly four fifths therefore comes from what participants believe others do and consider appropriate. Pre-treatment demographics account for 8.5%, technology attitudes for 5.9%, post-treatment demographics for 5.4%, and general attitudes

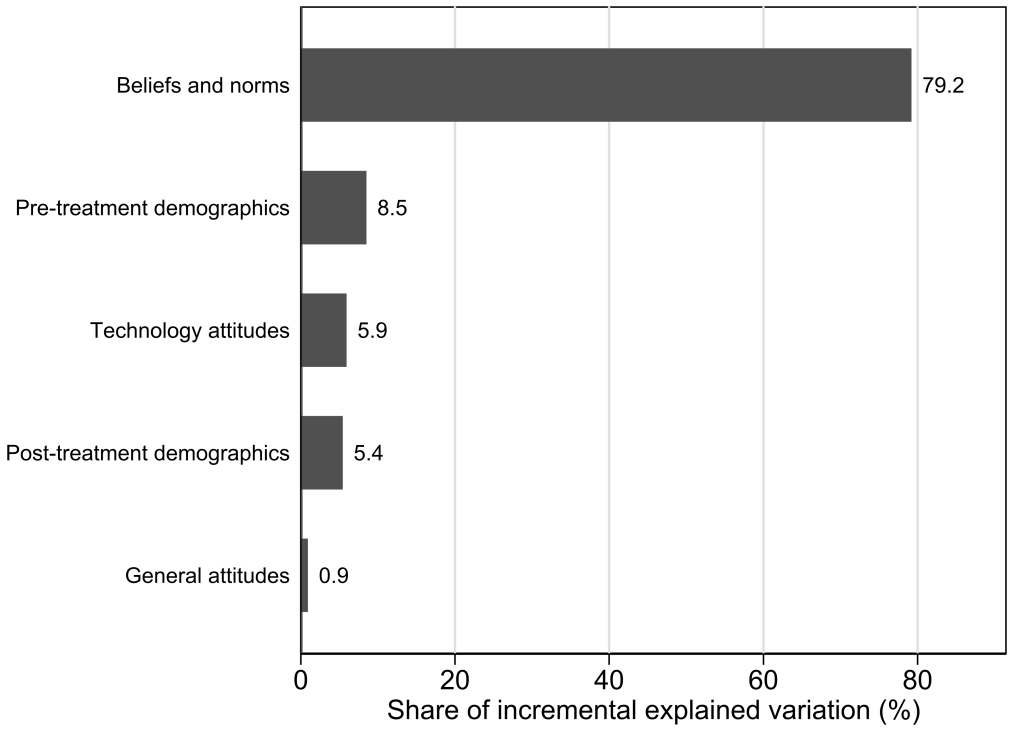


Figure 5: Observable contributions to explained variation in Follow

Note: Exact grouped Shapley–Owen allocation of the increase in ordinary R^2 from adding five covariate blocks among the three AI treatments. Treatment indicators and framing enter every coalition. All subset models use the full-model complete-case sample ($N = 1,198$). The R^2 rises from 0.016 with treatment and framing alone to 0.222 with all blocks; shares refer to this increment.

for 0.9%. Appendix Table B.6 reports the underlying contributions. This decomposition describes who follows the AI. It does not apportion the treatment effect, which persists throughout, among mechanisms. The concentration in beliefs and norms nonetheless motivates a direct examination of how the AI's role affects beliefs.

5.5 Potential Mechanisms

Our framework separates two channels through which the AI's initiative can reduce reliance: a lower weight on the assessment, $\lambda_I < \lambda_P, \lambda_A$, and a positive net procedural term, $c_I = \psi\phi_I - b_I > 0$, in which the value of authoring one's decision outweighs the benefits of following a preset action. If the penalty operated only through λ_I, conditioning on reported beliefs would absorb most of it. We examine the evidence for each channel and then evaluate two alternative explanations.

Participants' stated preferences mirror the agentic penalty. Figure 6 uses the Baseline treatment, whose participants had not been assigned an AI role and could therefore rate the three roles without having been assigned one. On a five-point scale, mean willingness to use prediction is 3.45, and willingness to use advice is 3.37. Willingness to delegate the decision is only 2.39. Both differences from delegation yield $p < 0.001$. The corresponding top-two response shares are 61.6%, 58.0%, and 23.1%. The aversion is specific to delegation, the role closest to Initiate, and it is expressed by participants who had not experienced the Initiate treatment. Baseline participants in our pilot study, two years earlier and with different payoffs, gave almost the same ratings: 3.45 for prediction, 3.19 for advice, and 2.48 for delegation (Appendix C). In the framework, these ratings speak to the value of authorship. Baseline participants never saw an assessment, so the weight on it plays no role in their ratings, and they prefer the roles that leave them the author of the decision ($\psi > 0$, with ϕ largest for delegation).

The assessment also moves beliefs less when the AI takes the initiative. Table 3 reports participants' post-choice expectations of Participant B's return rate. A positive rather than negative assessment raises the expected return rate by 18.3 percentage points in Predict, 14.1 points in Advise, and 6.7 points in Initiate. The Initiate response differs from Predict ($p = 0.001$) and Advise ($p = 0.026$). Average beliefs do not differ by treatment (Appendix B.9). The AI's role therefore does not shift what participants expect in general, but it changes how strongly the assessment informs those expectations. This pattern is consistent with a lower weight on the assessment under the AI's initiative, $\lambda_I < \lambda_P, \lambda_A$. Because beliefs were elicited after the choice, it is not a separately identified mediation effect.

Weaker belief responsiveness does not account for the whole penalty. The full specification in column (2) of Table 2 conditions on participants' reported beliefs and social-appropriateness judgements. If the penalty operated only through beliefs, the Predict–Initiate and Advise–Initiate

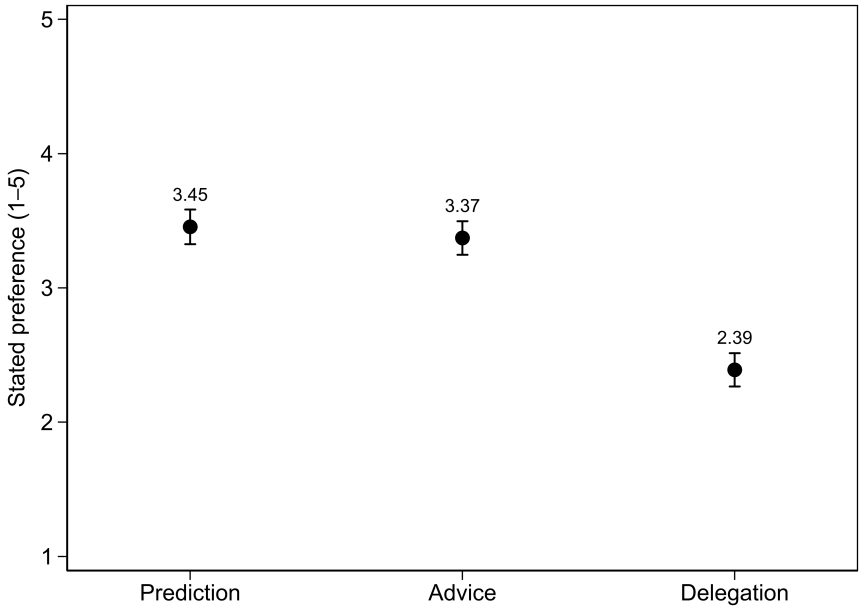


Figure 6: Baseline participants' stated preferences over AI roles

Note: Mean ratings with 95% confidence intervals among Baseline participants. Higher values indicate greater willingness to use the respective form of AI involvement.

differences in responsiveness should shrink towards zero. Instead, they are 13.5 and 16.6 percentage points, compared with 15.0 and 16.5 points in the pre-treatment benchmark. Participants who report the same beliefs and norms still rely less on the assessment when the AI has initiated the action. This persistence is consistent with a positive net procedural term, $c_I > 0$: the value of authorship outweighs the benefits of following a preset action. It is also consistent with the aversion to delegation that Baseline participants express.

Two alternative explanations find no support. Perceived classifier accuracy is virtually identical across roles: means in the three AI treatments range only from 3.56 to 3.58 on the five-point scale, and the all-treatment equality test yields $p = 0.981$. Participants did not override the AI's action because they judged its assessment to be worse. Nor is the penalty concentrated among participants with stronger trait reactance. The eleven-item scale is reliable ($\alpha = 0.875$), but continuous reactance does not differentially predict Follow across the three AI treatments (joint $p = 0.952$).

Outcome / test	Predict	Advise	Initiate	p
Belief response to positive assessment (pp)	18.3	14.1	6.7	
Predict = Initiate				0.001
Advise = Initiate				0.026
Perceived AI accuracy (1–5; all-treatment test)	3.58	3.56	3.57	0.981
Reactance effect on Follow (pp per scale point)	0.9	-0.3	0.4	0.952
Standardised scale reliability		$\alpha = 0.875$		

Note: Belief responses are minimally adjusted positive-minus-negative assessment differences with robust inference; beliefs were elicited after the trust decision. The perceived-accuracy row reports means in the three AI treatments and the Kruskal–Wallis test across all four treatments. Reactance entries are marginal effects from a linear probability model with continuous role interactions and framing; the reported p-value jointly tests the interactions.

Table 3: Potential mechanisms and alternative explanations

The evidence is therefore inconsistent with a general judgement that the classifier is less accurate when it takes the initiative and with trait reactance as the source of the effect. It is consistent with an account with two complementary parts. Giving the AI an agentic role weakens how strongly its information enters beliefs, and the persistence of the penalty is consistent with a cost of confirming an action the AI has initiated that reported beliefs and norms do not capture. In terms of the framework, the evidence points to both a lower λ_I and $c_I > 0$. Because $b_I \geq 0$, the value of authorship $\psi\phi_I$ exceeds b_I: whatever status-quo bias or responsibility shifting favoured confirmation was outweighed by the value participants placed on authoring their decision.

Result 3. *The agentic penalty is accompanied by weaker belief responsiveness to the assessment and persists conditional on reported beliefs and norms. It is not explained by perceived classifier accuracy or trait reactance.*

In sum, AI initiative causes lower following. The agentic penalty is symmetric across assessment direction, persists across framing and specifications, and is accompanied by weaker belief responsiveness rather than by lower perceived accuracy or stronger trait reactance. The Discussion turns to what this implies.

6 Discussion

Our experiment reveals two interrelated behavioural findings. First, following is lower when the AI takes the initiative than when it supplies a prediction or advice. Second, participants' trust responds less sharply to the classifier assessment in Initiate than in Predict or Advise, and it does so symmetrically for assessments that encourage and discourage trust. Both patterns are consistent with a

value of retaining the initiative. The continuous-reactance analysis, however, provides no evidence that the treatment difference is concentrated among participants with higher trait reactance. The value of the initiative is therefore not confined to participants with a particular personality trait.

Initiative, authority, and following. Predict and Advise produce almost identical rates, while both differ from Initiate. This pattern is consistent with the hypothesis that individuals value making their own decisions and are willing to depart from the AI's action even when override is costless. The information transmitted to Participant A is identical across all three AI treatments. The same underlying assessment drives the prediction, the recommendation, and the AI-initiated action. The observed differences in following therefore cannot be attributed to differences in informational content. Instead, they reflect the institutional role of the AI. The design also separates two components of that role. Final authority is held fixed: participants in every treatment make the same consequential choice and can override the AI at no cost. What varies is who initiates the action. The agentic penalty thus shows that people respond to the initiative, not only to the authority.

This interpretation aligns with the decision-rights literature pioneered by Bartling et al. (2014), the control-premium evidence of Owens et al. (2014), and the distinction between initiating and implementing decisions in organisations (Sloof and von Siemens, 2015). Our contribution is to show that this mechanism extends to a setting where the decision-maker interacts with a human counterpart under strategic uncertainty rather than making an individual choice. The source of the reallocation is an AI rather than another person. Only the initiative, not the final authority, is reallocated. An AI allows the design to hold the information and its disclosed accuracy exactly fixed across roles; a human expert in the same roles would add intentions, a stake, and blame, and thus a third person to the trust problem. Whether the penalty is specific to AI or a general response to delegated initiative is a natural extension, using a human-expert version of the same treatments.

Why status-quo bias does not prevail. The direction of the effect is informative because two well-documented forces predicted the opposite. An action already initiated by the AI resembles a default, and defaults reliably raise adherence through inertia, status-quo bias, and implicit recommendation (Samuelson and Zeckhauser, 1988; Madrian and Shea, 2001; Johnson and Goldstein, 2003; McKenzie et al., 2006; Jachimowicz et al., 2019). Delegation also allows people to shift responsibility for outcomes that depend on others (Bartling and Fischbacher, 2012; Hamman et al., 2010), which should make acting on the AI's assessment easier precisely when trusting exposes the participant to betrayal (Bohnet and Zeckhauser, 2004; Bohnet et al., 2008). In our data, neither force prevails. The Initiate treatment required active confirmation rather than relying on a passive default, which removes inattention as a channel but leaves status-quo bias and the implicit recommendation intact. Had status-quo bias dominated, following would have risen in both directions.

Had responsibility shifting dominated, trust would have risen mainly after positive assessments. The symmetric reduction in responsiveness we observe fits neither pattern. The value of retaining the initiative outweighs whatever status-quo bias and responsibility shifting favoured confirmation.

Symmetric reduction in responsiveness and the content of the assessment. The symmetry also clarifies whether the AI's role interacts with the content of its assessment. Prior evidence shows that the effect of AI involvement can depend on the moral or social content of the choice (Chugunova and Sele, 2022; March, 2021). One might therefore have expected the AI's initiative to matter differently for an assessment that encourages trust in a stranger and for one that warns against it. It does not. The AI's initiative weakens both the encouragement and the warning by similar amounts. The agentic penalty is a property of the AI's role, not of the content of its assessment. For applications, this means that AI initiative does not simply make people more or less trusting. It makes them less responsive to what the AI knows, in whichever direction that knowledge points.

Reactance as a mechanism. If the lower Follow rate under Initiate reflects a preference for autonomy, one might expect it to be amplified among participants with higher trait reactance, a trait that captures the tendency to resist perceived restrictions on freedom of choice (Brehm, 1966; Brehm and Brehm, 1981). We do not observe that pattern. The continuous treatment-by-reactance interactions are jointly null. The behavioural effect remains consistent with a decision-rights interpretation, but the trait evidence does not identify reactance as its mechanism. The value of decision rights documented in incentivised experiments is not a trait measure of reactance, so the null does not speak against a procedural cost of ceding the initiative. It provides no evidence that this cost is concentrated among participants with the highest reactance scores.[7]

Language frame. A distinctive feature of our design is the variation in how the trust game is described. Participants faced either the neutral frame ("send," "keep," "return") or the natural trust frame ("trust," "distrust," "be trustworthy"). One might expect explicit trust language to raise the moral salience of the decision and thereby interact with the AI's role. For example, a recommendation to "trust" may carry more normative force than a recommendation to "send." The frame does matter for the level of reliance: in the three AI treatments, the natural trust frame raises following by 5.2 percentage points. It does not, however, change the agentic penalty. Our treatment-by-framing interaction tests reveal no statistically significant interaction effects for either following or trust, and the ordering Predict $\approx$ Advise $>$ Initiate appears in both frames. Trust

[7] A post-game item asking participants in the AI treatments whether they would have chosen differently without the AI tool was not recorded owing to an implementation error in the survey module. We therefore cannot corroborate the procedural-cost interpretation with this self-report.

language thus raises reliance on the AI across the board, while the agentic penalty depends on whether the AI predicts, advises, or initiates the action. This finding also makes it less likely that the penalty is an artefact of the wording of the decision.

Beliefs and the two channels. The AI's role does not shift average beliefs about others' behaviour or about the social appropriateness of trusting and returning (Appendix B.9). It does change how strongly the assessment informs those beliefs. The positive-minus-negative difference in expected return rates is 18.3 percentage points in Predict and 14.1 points in Advise, but only 6.7 points in Initiate. At the same time, the penalty in trust responsiveness remains of similar size when we condition on reported beliefs and norms. Taken together, these facts are consistent with the two channels of the framework. The AI's initiative lowers the weight placed on the assessment in belief formation, and the persistence of the penalty is consistent with a cost of confirming an AI-initiated action that reported beliefs do not capture. Because beliefs were elicited after the choice, we present this as a pattern consistent with the framework rather than as a formal mediation result. Separating the two channels experimentally, for example by eliciting incentivised beliefs before the choice or by varying the cost of override, is a natural next step.

Classifier performance in the experimental sample. The classifier's assessments had little discriminatory power for counterparts' return decisions in our sample (Appendix B.12). Participants could not know this. They did not observe counterpart characteristics, were told the same held-out accuracy in all three AI treatments, and received no feedback before the questionnaire. The treatment comparisons are therefore unaffected, and they measure how the AI's role changes reliance on information that participants believed to be highly accurate. This feature also clarifies the interpretation of overrides: participants in Initiate did not override the AI because they had detected its errors. We draw no conclusions about whether following the assessment improved participants' earnings. Whether more or less reliance is desirable depends on the system's accuracy in the population where it is deployed, a question our design deliberately holds outside participants' knowledge.

Returning behaviour. Participants made the Participant B decision after experiencing, as Participant A, the same AI role as their counterpart, but without learning which assessment their counterpart had received. Treatment could therefore affect return decisions through this experience, for example because participants in the AI treatments knew that an assessment of their own behaviour might have been shown to their counterpart. Return rates vary modestly across treatments, with somewhat higher returning in Predict than in Advise and Initiate in the pre-treatment benchmark. The pattern is not stable after sequential inclusion of covariates and receives no mechanism

interpretation (Appendix B.10).

Implications for the deployment of AI in interpersonal decisions. In our setting, moving the AI from prediction or advice to initiative forfeits about a quarter of its influence on behaviour, although its information, the available actions, and the participant's final authority are unchanged and override is costless. The lesson for systems that assess one person and predict, advise, or act for another is that the deployment role governs how much the system's information is used. Counterparty screening, candidate assessment, and automated investment are applications in which the same shift from prediction or advice to initiative occurs, although our evidence comes from an incentivised human–human trust interaction rather than from these environments. Designers who assume that the information of a system that takes the initiative will be used at least as much as that of an advising system will overstate its influence. Where the objective is for people to act on an accurate assessment, prediction and advice achieve more of it. Where the objective is to preserve human judgement, giving the AI the initiative does so, but at the cost of information use. Preserving the user's final authority is not sufficient to avoid this trade-off. The AI agentic penalty arises even when the person can override the AI at no cost. In the field, agentic systems also save their users effort, a benefit our design removes by requiring the same single choice in every treatment. The agentic penalty is therefore a cost that such workload savings must outweigh.

7 Conclusion

This paper investigates how the institutional role of an AI system shapes reliance on its information in an incentivised human–human trust environment. We compare an AI that predicts, advises, or initiates the decision while holding the AI's assessment and the participant's consequential choices fixed. Our preregistered experiment reveals a clear and robust pattern. Following the AI is 8–9 percentage points lower when the system takes the initiative than when it predicts or recommends, although the information transmitted is identical and override is costless. Prediction and advice are indistinguishable. Measured against a Baseline without access to the assessment, the AI's initiative forfeits about one quarter of its influence on behaviour. We call this reduction the AI agentic penalty.

Beyond the aggregate result, the penalty is symmetric in the direction of the assessment. The AI's initiative weakens the encouragement to trust and the warning against it by similar magnitudes. This distinguishes the penalty from a general shift towards more or less trust and from accounts in which the initiative matters mainly through one direction, such as responsibility shifting when the AI extends trust on the participant's behalf. It also runs counter to status-quo bias in favour of an action that has already been initiated. Whatever status-quo bias or implicit recom-

mendation the AI's action carried was outweighed by the value participants placed on the initiative.

The mechanism evidence is consistent with two complementary channels. The AI's initiative weakens how strongly the assessment moves participants' beliefs about their counterpart, yet the penalty remains similar after conditioning on reported beliefs and norms. Baseline participants, who were not assigned an AI role, express a specific aversion to delegation but not to prediction or advice. Perceived accuracy is identical across roles, and the penalty is not concentrated among participants with high trait reactance. The decision-rights literature has established that people value holding the final authority. Our results show that authority is not the whole story. People also respond to who initiates the action, and preserving final authority does not make AI initiative equivalent to prediction or advice.

Our results carry implications for the deployment of AI in decisions that involve trust between people. In this setting, the role assigned to the system, and not only the accuracy of its assessment, governs how much its information is used. Deploying the system to predict or advise preserves its influence. Giving it the initiative, even with costless override, forfeits part of that influence in both directions. Applications in counterparty screening, candidate assessment, and automated financial decisions share the structure of our experiment, but our evidence comes from a one-shot, incentivised human–human trust interaction, and we present these settings as applications rather than as tested environments.

Several extensions follow directly from the design. Repeated interaction with feedback would show whether the agentic penalty persists once people observe the system's successes and mistakes. Varying the cost of override, eliciting incentivised beliefs before the choice, and recording decision processes would separate a lower weight on the assessment from the procedural cost of confirming an AI-initiated action. A treatment in which the same action is pre-selected without the AI initiating it would separate the pull of a default from the response to the AI's initiative. Each of these steps builds on the result established here: in interpersonal trust, an AI loses influence when it gains the initiative.

References

AGARWAL, R. AND J. PRASAD (1998): "A Conceptual and Operational Definition of Personal Innovativeness in the Domain of Information Technology," *Information Systems Research*, 9, 204–215.

AGRAWAL, A., J. GANS, AND A. GOLDFARB (2018): *Prediction Machines: The Simple Economics of Artificial Intelligence*, Harvard Business Press.

ANGELOVA, V., W. DOBBIE, AND C. S. YANG (2026): "Algorithmic Recommendations and Human Discretion," *Review of Economic Studies*, 93, 2250–2283.

BABIN, J. J. (2020): "Linguistic signaling, emojis, and skin tone in trust games," *PLoS One*, 15, e0233277.

BARTLING, B., E. FEHR, AND H. HERZ (2014): "The Intrinsic Value of Decision Rights," *Econometrica*, 82, 2005–2039.

BARTLING, B. AND U. FISCHBACHER (2012): "Shifting the Blame: On Delegation and Responsibility," *Review of Economic Studies*, 79, 67–87.

BERGER, B., M. ADAM, A. RÜHR, AND A. BENLIAN (2021): "Watch Me Improve—Algorithm Aversion and Demonstrating the Ability to Learn," *Business & Information Systems Engineering*, 63, 55–68.

BOHNET, I., F. GREIG, B. HERRMANN, AND R. ZECKHAUSER (2008): "Betrayal Aversion: Evidence from Brazil, China, Oman, Switzerland, Turkey, and the United States," *American Economic Review*, 98, 294–310.

BOHNET, I. AND R. ZECKHAUSER (2004): "Trust, risk and betrayal," *Journal of Economic Behavior & Organization*, 55, 467–484.

BREHM, J. W. (1966): *A Theory of Psychological Reactance*, Academic Press.

BREHM, S. AND J. W. BREHM (1981): "Psychological Reactance: A Theory of Freedom and Control," *Academic Press*.

CASTELO, N., M. W. BOS, AND D. R. LEHMANN (2019): "Task-dependent algorithm aversion," *Journal of Marketing Research*, 56, 809–825.

CHEN, D. L., M. SCHONGER, AND C. WICKENS (2016): "oTree—An open-source platform for laboratory, online, and field experiments," *Journal of Behavioral and Experimental Finance*, 9, 88–97.

CHETTY, R., A. HOFMEYR, H. KINCAID, AND B. MONROE (2021): "The trust game does not (only) measure trust: The risk-trust confound revisited," *Journal of Behavioral and Experimental Economics*, 90, 101520.

CHOI, G. S. AND V. H. STORR (2020): "Market interactions, trust and reciprocity," *PloS one*, 15, e0232704.

CHUGUNOVA, M. AND D. SELE (2022): "We and It: An Interdisciplinary Review of the Experimental Evidence on How Humans Interact with Machines," *Journal of Behavioral and Experimental Economics*, 99, 101897.

COFFMAN, L. C. (2011): "Intermediation Reduces Punishment (and Reward)," *American Economic Journal: Microeconomics*, 3, 77–106.

DIETVORST, B. J., J. P. SIMMONS, AND C. MASSEY (2015): "Algorithm aversion: people erroneously avoid algorithms after seeing them err." *Journal of Experimental Psychology: General*, 144, 114.

——— (2018): "Overcoming Algorithm Aversion: People Will Use Imperfect Algorithms If They Can (Even Slightly) Modify Them," *Management Science*, 64, 1155–1170.

DOHMEN, T., A. FALK, D. HUFFMAN, U. SUNDE, J. SCHUPP, AND G. G. WAGNER (2011): "Individual Risk Attitudes: Measurement, Determinants, and Behavioral Consequences," *Journal of the European Economic Association*, 9, 522–550.

ENDSLEY, M. R. AND E. O. KIRIS (1995): "The Out-of-the-Loop Performance Problem and Level of Control in Automation," *Human Factors*, 37, 381–394.

EUROPEAN PARLIAMENT AND COUNCIL OF THE EUROPEAN UNION (2024): "Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)," Official Journal of the European Union, L 2024/1689, 12 July 2024.

FALK, A., A. BECKER, T. DOHMEN, B. ENKE, D. HUFFMAN, AND U. SUNDE (2018): "Global Evidence on Economic Preferences," *Quarterly Journal of Economics*, 133, 1645–1692.

FALK, A. AND M. KOSFELD (2006): "The hidden costs of control," *American Economic Review*, 96, 1611–1630.

FEHR, E., H. HERZ, AND T. WILKENING (2013): "The Lure of Authority: Motivation and Incentive Effects of Power," *American Economic Review*, 103, 1325–1359.

FREY, B. S., M. BENZ, AND A. STUTZER (2004): "Introducing Procedural Utility: Not Only What, but Also How Matters," *Journal of Institutional and Theoretical Economics (JITE)*, 160, 377–401.

FUSTER, A., P. GOLDSMITH-PINKHAM, T. RAMADORAI, AND A. WALTHER (2022): "Predictably Unequal? The Effects of Machine Learning on Credit Markets," *Journal of Finance*, 77, 5–47.

GEFEN, D. AND D. W. STRAUB (2004): "Consumer Trust in B2C E-Commerce and the Importance of Social Presence: Experiments in E-Products and E-Services," *Omega*, 32, 407–424.

GERMANN, M. AND C. MERKLE (2023): "Algorithm Aversion in Delegated Investing," *Journal of Business Economics*, 93, 1691–1727.

GREEVINK, I., T. OFFERMAN, AND G. ROMAGNOLI (2026): "AI-Powered Promises: The Influence of ChatGPT on Trust and Trustworthiness," *Journal of Economic Behavior & Organization*, 246, 107538.

HAMMAN, J. R., G. LOEWENSTEIN, AND R. A. WEBER (2010): "Self-Interest through Delegation: An Additional Rationale for the Principal–Agent Relationship," *American Economic Review*, 100, 1826–1846.

HARMS, J., K. LIKET, J. PROTZKO, AND V. SCHÖLMERICH (2017): "Free to help? An experiment on free will belief and altruism," *PLoS ONE*, 12.

HEYES, A. AND J. A. LIST (2016): "Supply and demand for discrimination: Strategic revelation of own characteristics in a trust game," *American Economic Review*, 106, 319–323.

HIEPLER, B., J. SCHMITZ, F. SCHNEIDER, AND S. ZEISBERGER (2026): "Delegation to Humans and Algorithms in Financial Decision Making," Tech. Rep. AP 2026-007, Network for Studies on Pensions, Aging and Retirement (Netspar).

HOFFMAN, M., L. B. KAHN, AND D. LI (2018): "Discretion in Hiring," *Quarterly Journal of Economics*, 133, 765–800.

HOLZMEISTER, F., M. HOLMÉN, M. KIRCHLER, M. STEFAN, AND E. WENGSTRÖM (2023): "Delegation Decisions in Finance," *Management Science*, 69, 4828–4844.

HONG, S.-M. AND S. FAEDDA (1996): "Refinement of the Hong Psychological Reactance Scale," *Educational and Psychological Measurement*, 56, 173–182.

IVANOVA-STENZEL, R. AND M. TOLKSDORF (2024): "Measuring Preferences for Algorithms—How Willing Are People to Cede Control to Algorithms?" *Journal of Behavioral and Experimental Economics*, 112, 102270.

——— (2025): "Delegating in the Age of AI: Preferences for Decision Autonomy," Tech. Rep. Discussion Paper No. 558, Collaborative Research Center Transregio 190, Rationality and Competition.

JACHIMOWICZ, J. M., S. DUNCAN, E. U. WEBER, AND E. J. JOHNSON (2019): "When and Why Defaults Influence Decisions: A Meta-Analysis of Default Effects," *Behavioural Public Policy*, 3, 159–186.

JOHNSON, E. J. AND D. GOLDSTEIN (2003): "Do Defaults Save Lives?" *Science*, 302, 1338–1339.

KEE, J., M. KNUTH, J. N. LAHEY, AND M. A. PALMA (2021): "Does eye-tracking have an effect on economic behavior?" *Plos one*, 16, e0254867.

KLEINBERG, J., H. LAKKARAJU, J. LESKOVEC, J. LUDWIG, AND S. MULLAINATHAN (2018): "Human Decisions and Machine Predictions," *Quarterly Journal of Economics*, 133, 237–293.

KOVALEVA, A. (2012): *The IE-4: Construction and Validation of a Short Scale for the Assessment of Locus of Control*, vol. 9 of *GESIS-Schriftenreihe*, Cologne: GESIS – Leibniz Institute for the Social Sciences.

KRUPKA, E. L. AND R. A. WEBER (2013): "Identifying Social Norms Using Coordination Games: Why Does Dictator Game Sharing Vary?" *Journal of the European Economic Association*, 11, 495–524.

LI, Z., P.-H. LIN, S.-Y. KONG, D. WANG, AND J. DUFFY (2021): "Conducting large, repeated, multi-game economic experiments using mobile platforms," *PloS one*, 16, e0250668.

LOGG, J. M., J. A. MINSON, AND D. A. MOORE (2019): "Algorithm appreciation: People prefer algorithmic to human judgment," *Organizational Behavior and Human Decision Processes*, 151, 90–103.

MADRIAN, B. C. AND D. F. SHEA (2001): "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior," *Quarterly Journal of Economics*, 116, 1149–1187.

MARCH, C. (2021): "Strategic interactions between humans and artificial intelligence: Lessons from experiments with computer players," *Journal of Economic Psychology*, 87, 102426.

MCKENZIE, C. R. M., M. J. LIERSCH, AND S. R. FINKELSTEIN (2006): "Recommendations Implicit in Policy Defaults," *Psychological Science*, 17, 414–420.

NISHINA, K., H. TAKAGISHI, M. INOUE-MURAYAMA, H. TAKAHASHI, AND T. YAMAGISHI (2015): "Polymorphism of the oxytocin receptor gene modulates behavioral and attitudinal trust among men but not women," *PloS one*, 10, e0137089.

NOUSSAIR, C. N. AND J. STOOP (2015): "Time as a medium of reward in three social preference experiments," *Experimental Economics*, 18, 442–456.

ONNASCH, L., C. D. WICKENS, H. LI, AND D. MANZEY (2014): "Human Performance Consequences of Stages and Levels of Automation: An Integrated Meta-Analysis," *Human Factors*, 56, 476–488.

OWENS, D., Z. GROSSMAN, AND R. FACKLER (2014): "The Control Premium: A Preference for Payoff Autonomy," *American Economic Journal: Microeconomics*, 6, 138–161.

ÖZER, Ö., U. SUBRAMANIAN, AND Y. WANG (2018): "Information Sharing, Advice Provision, or Delegation: What Leads to Higher Trust and Trustworthiness?" *Management Science*, 64, 474–493.

PALAN, S. AND C. SCHITTER (2018): "Prolific. ac—A subject pool for online experiments," *Journal of behavioral and experimental finance*, 17, 22–27.

POWELL, P. A., O. WILLS, G. REYNOLDS, K. PUUSTINEN-HOPPER, AND J. ROBERTS (2018): "The effects of exposure to images of others' suffering and vulnerability on altruistic, trust-based, and reciprocated economic decision-making," *Plos one*, 13, e0194569.

PRAHL, A. AND L. VAN SWOL (2017): "Understanding algorithm aversion: When is advice from automation discounted?" *Journal of Forecasting*, 36, 691–702.

SAMUELSON, W. AND R. ZECKHAUSER (1988): "Status Quo Bias in Decision Making," *Journal of Risk and Uncertainty*, 1, 7–59.

SELTEN, R. (1967): "Die strategiemethode zur erforschung des eingeschr nkt rationale verhaltens im rahmen eines oligopolexperiments," *Beitr ge zur experimentellen Wirtschaftsforschung*, 136.

SHEN, C., J. KARPUS, T. KOSCH, D. FERNANDES, B. MELLO, R. WELSCH, AND S. VILLA (2025): "The Impact of Asymmetric AI Assistance on Decision-Making in Social Dilemmas: A Study on Human Augmentation in Economic Games," in *Proceedings of the Augmented Humans International Conference 2025*, 187–198.

SINDERMANN, C., P. SHA, M. ZHOU, J. WERNICKE, H. S. SCHMITT, M. LI, R. SARIYSKA, M. STAVROU, B. BECKER, AND C. MONTAG (2021): "Assessing the Attitude Towards Artificial Intelligence: Introduction of a Short Measure in German, Chinese, and English Language," *KI – Künstliche Intelligenz*, 35, 109–118.

SLOOF, R. AND F. A. VON SIEMENS (2015): "Decision Initiation, Decision Implementation, and the Allocation of Decision Rights," Tech. Rep. Discussion Paper 15-105/VII, Tinbergen Institute.

VAN KLINGEREN, F. (2020): "Playing nice in the sandbox: On the role of heterogeneity, trust and cooperation in common-pool resources," *Plos one*, 15, e0237870.

ZHANG, L., I. PENTINA, AND Y. FAN (2021): "Who do you choose? Comparing perceptions of human vs robo-advisor in the context of financial services," *Journal of Services Marketing*.

Appendix

This appendix documents the preregistration map, full specifications, robustness checks, classifier performance in the experimental sample, the pilot study, and additional analyses supporting the main text.

A Datasets Used to Train the AI Tool

Publication	Observations
Nishina et al. (2015)	380
Noussair and Stoop (2015)	44
Heyes and List (2016)	680
Powell et al. (2018)	5,230
Babin (2020)	43
Choi and Storr (2020)	200
van Klingeren (2020)	10,440
Chetty et al. (2021)	2,160
Kee et al. (2021)	138
Li et al. (2021)	442
Pilot study (Appendix C)	266
Total	20,023

Table A.1: Experimental datasets used to train the AI tool

Note: Observations are return decisions. After one observation with a missing outcome was dropped, the classifier was trained on 16,017 observations and evaluated on a hold-out sample of 4,005, on which its accuracy was 89.86%. The pilot study contributes the return decisions of all its sessions.

B Additional Statistics and Results

B.1 Preregistration and Multiplicity

The preregistered power calculation defined a five-treatment-comparison family: Predict–Baseline, Advise–Baseline, Initiate–Baseline, Predict–Advise, and Advise–Initiate. Table B.1 reports those two-sided Boschloo tests. The Predict–Initiate treatment comparison is additionally reported in Table 1 because it is explicitly part of preregistered Hypothesis 1, although it is not among those five treatment comparisons. At the Bonferroni-adjusted threshold of $\alpha = 0.01$, Predict–Advise is not rejected, whereas Advise–Initiate is rejected.

B.2 Balance and Sample Description

Table B.2 reports pre-treatment covariates by experimental condition together with Kruskal–Wallis p-values for cross-treatment differences. Age differs across treatments ($p = 0.024$), while adults

Treatment comparison	Difference (pp)	p
Predict–Baseline	34.0	< 0.001
Advise–Baseline	34.6	< 0.001
Initiate–Baseline	25.8	< 0.001
Predict–Advise	-0.7	0.849
Advise–Initiate	8.9	0.002

Note: Differences are first-listed treatment minus second-listed treatment. Two-sided Boschloo tests use the ordinary two-sided Fisher exact p-value as the ordering statistic. For Baseline, the assessment was generated but not shown to participants, so the quantity is alignment with the AI assessment rather than following.

Table B.1: Five-treatment-comparison family used in the preregistered power calculation

per household shows weaker evidence of a difference ($p = 0.093$). Both differences are small in magnitude. The largest pairwise standardised mean differences are 0.179 for age and 0.165 for adults per household. More importantly, an omnibus multinomial-logit test does not reject the joint null that the full set of pre-treatment covariates is unrelated to treatment assignment ($\chi^2(69) = 65.82$, $p = 0.586$, $N = 1,596$). The balance evidence is therefore consistent with successful random assignment.

B.3 Complete Follow Probit Tables

Tables B.3–B.5 present the complete coefficient estimates for the stepwise probit regressions of Follow. Column (1) includes the AI-role treatment indicators and the randomised framing indicator. Column (2) adds pre-treatment demographics and serves as the pre-treatment benchmark. Columns (3)–(6) then add post-treatment demographics, technology attitudes, beliefs and norms, and general attitudes sequentially, with column (6) reported as the full specification. The ordering Predict $\approx$ Advise $>$ Initiate is stable after sequential inclusion of covariates, with no material change in direction or statistical inference relative to column (2).

B.4 Observable Heterogeneity Decomposition

Table B.6 reports the grouped Shapley–Owen decomposition underlying Figure 5. The decomposition allocates the increase in ordinary R^2 from adding five covariate blocks to a linear probability model of Follow among the three AI treatments. Treatment and framing enter every coalition, and all subset models use the full-model complete-case sample of 1,198 participants. The bottom rows report the R^2 with treatment and framing alone, the full-model R^2, and the increment that the shares divide.

B.5 Complete Trust-Calibration Specifications

Table B.7 reports the complete set of positive-minus-negative assessment differences: raw linear probability, framing-adjusted, pre-treatment benchmark, full specification, and pair-clustered

	Baseline	Predict	Advise	Initiate	Total	p-value
Age	43.90	43.57	42.49	41.51	42.87	$p = 0.024$
	(13.29)	(13.15)	(12.74)	(13.39)	(13.17)	
Female	0.523	0.454	0.511	0.495	0.496	$p = 0.225$
	(0.500)	(0.498)	(0.500)	(0.501)	(0.500)	
Continent	4.940	4.942	4.970	4.967	4.955	$p = 0.639$
	(0.456)	(0.436)	(0.306)	(0.328)	(0.387)	
Education	3.570	3.704	3.675	3.606	3.639	$p = 0.566$
	(1.231)	(1.164)	(1.210)	(1.210)	(1.204)	
Working	0.794	0.777	0.826	0.798	0.799	$p = 0.370$
	(0.405)	(0.417)	(0.379)	(0.402)	(0.401)	
Student	0.0377	0.0376	0.0347	0.0455	0.0388	$p = 0.879$
	(0.191)	(0.190)	(0.183)	(0.209)	(0.193)	
Economics/Business	0.206	0.228	0.233	0.237	0.226	$p = 0.722$
administration background	(0.405)	(0.420)	(0.423)	(0.426)	(0.418)	
Non-White	0.131	0.140	0.151	0.172	0.148	$p = 0.403$
	(0.337)	(0.348)	(0.359)	(0.378)	(0.356)	
Income	1.869	1.935	2.007	1.939	1.938	$p = 0.230$
	(0.930)	(0.959)	(0.994)	(0.939)	(0.956)	
Born in England	0.759	0.744	0.787	0.790	0.770	$p = 0.345$
	(0.428)	(0.437)	(0.410)	(0.408)	(0.421)	
Religion	0.510	0.521	0.538	0.548	0.529	$p = 0.814$
	(0.622)	(0.625)	(0.607)	(0.648)	(0.625)	
Adults in household	2.088	2.078	2.238	2.215	2.155	$p = 0.093$
	(0.881)	(0.751)	(1.640)	(0.898)	(1.103)	
Children in household	0.608	0.662	0.680	0.755	0.676	$p = 0.244$
	(0.891)	(1.024)	(0.946)	(1.042)	(0.978)	

Note: Columns report treatment-specific and full-sample means, with standard deviations in parentheses. The final column reports p-values from two-sided Kruskal–Wallis tests across the four treatments.

Table B.2: Descriptive statistics by treatment

Dep. Var.: Follow	(1)	(2)	(3)	(4)	(5)	(6)
Predict	0.976***	0.996***	1.003***	1.010***	1.033***	1.026***
	(0.098)	(0.098)	(0.099)	(0.100)	(0.108)	(0.108)
Advise	1.009***	1.023***	1.026***	1.047***	0.999***	1.002***
	(0.097)	(0.098)	(0.099)	(0.100)	(0.105)	(0.105)
Initiate	0.693***	0.710***	0.711***	0.720***	0.545***	0.548***
	(0.093)	(0.094)	(0.094)	(0.095)	(0.108)	(0.108)
Natural trust	0.196***	0.187***	0.192***	0.182***	0.139*	0.144*
	(0.069)	(0.070)	(0.070)	(0.071)	(0.076)	(0.077)
Age		0.000	-0.000	-0.002	-0.002	-0.002
		(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Female		0.085	0.093	0.108	0.041	0.056
		(0.071)	(0.074)	(0.077)	(0.080)	(0.081)
High school		0.103	0.112	0.059	0.074	0.077
		(0.194)	(0.195)	(0.194)	(0.205)	(0.205)
Attended college		-0.179	-0.184	-0.235	-0.176	-0.178
		(0.204)	(0.205)	(0.204)	(0.216)	(0.216)
Undergraduate degree		0.032	0.047	0.016	0.036	0.039
		(0.187)	(0.190)	(0.189)	(0.200)	(0.201)
Graduate degree or higher		-0.224	-0.191	-0.216	-0.123	-0.121
		(0.198)	(0.202)	(0.201)	(0.211)	(0.212)
Doctorate degree		0.168	0.235	0.189	0.237	0.246
		(0.266)	(0.275)	(0.274)	(0.282)	(0.282)
Working		-0.021	-0.001	-0.027	0.033	0.032
		(0.094)	(0.102)	(0.103)	(0.107)	(0.108)
Student		-0.099	-0.106	-0.116	-0.157	-0.139
		(0.193)	(0.192)	(0.193)	(0.198)	(0.200)
Economics/Business		0.133	0.157*	0.142	0.063	0.057
administration background		(0.085)	(0.086)	(0.088)	(0.090)	(0.090)
Non-White		-0.109	-0.074	-0.075	-0.110	-0.132
		(0.102)	(0.108)	(0.110)	(0.110)	(0.112)
£25.001 - £50.000			-0.144	-0.180**	-0.251***	-0.251***
			(0.090)	(0.092)	(0.096)	(0.096)
Pseudo R^2	0.084	0.094	0.101	0.113	0.182	0.183
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.3: Probit regressions of Follow (complete table, Part 1/3)

Dep. Var.: Follow	(1)	(2)	(3)	(4)	(5)	(6)
£50.001 - £75.000			0.015	-0.011	-0.110	-0.118
			(0.122)	(0.126)	(0.129)	(0.129)
£75.001 - £100.000			0.101	0.106	0.163	0.156
			(0.221)	(0.223)	(0.237)	(0.238)
More than £100.000			-0.134	-0.156	-0.252	-0.256
			(0.230)	(0.235)	(0.244)	(0.243)
Born in England			0.168**	0.171**	0.152*	0.152*
			(0.084)	(0.085)	(0.088)	(0.088)
Religious			-0.067	-0.082	-0.117	-0.115
			(0.079)	(0.080)	(0.084)	(0.084)
Other			-0.276**	-0.321**	-0.350**	-0.348**
			(0.140)	(0.140)	(0.147)	(0.148)
Adults in household			-0.015	-0.016	-0.034	-0.033
			(0.029)	(0.029)	(0.030)	(0.030)
Children in household			0.026	0.006	-0.002	-0.002
			(0.037)	(0.038)	(0.038)	(0.039)
Experience with AI				-0.142***	-0.112**	-0.119**
				(0.053)	(0.054)	(0.055)
Interaction with AI				-0.141***	-0.078	-0.077
				(0.048)	(0.050)	(0.050)
Profession with AI				-0.122	-0.144	-0.141
				(0.122)	(0.123)	(0.123)
Attitude to AI				0.129**	0.062	0.066
				(0.054)	(0.059)	(0.059)
Attitude to IT				-0.001	-0.059	-0.066
				(0.051)	(0.054)	(0.056)
Locus of control				-0.024	-0.049	-0.055
				(0.077)	(0.080)	(0.080)
Reactance				0.048	0.042	0.023
				(0.053)	(0.057)	(0.059)
Belief in free will				-0.018	-0.005	-0.004
				(0.019)	(0.020)	(0.020)
Pseudo R^2	0.084	0.094	0.101	0.113	0.182	0.183
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.4: Probit regressions of Follow (complete table, Part 2/3)

Dep. Var.: Follow	(1)	(2)	(3)	(4)	(5)	(6)
(Would) consider prediction					0.101**	0.101**
					(0.045)	(0.045)
(Would) consider					0.184***	0.189***
recommendation					(0.045)	(0.046)
(Would) consider delegation					0.148***	0.147***
					(0.036)	(0.036)
Belief that Participants A					-0.003	-0.003
send (0-100)					(0.002)	(0.002)
Belief that Participants B					0.003	0.003
return (0-100)					(0.002)	(0.002)
Appropriateness of					0.048	0.050
Participant A sending (1-4)					(0.081)	(0.081)
Appropriateness of					-0.027	-0.033
Participant A not sending (1-4)					(0.071)	(0.072)
Appropriateness of					-0.097	-0.094
Participant B returning (1-4)					(0.088)	(0.087)
Appropriateness of					0.003	0.004
Participant B not returning (1-4)					(0.068)	(0.068)
Willingness to take risks						0.015
						(0.018)
Trusting disposition						-0.046
						(0.046)
Altruism						0.000
						(0.001)
Social image concerns						-0.004
						(0.017)
Constant	-0.129*	-0.145	-0.188	0.456	-0.351	-0.209
	(0.072)	(0.267)	(0.282)	(0.511)	(0.620)	(0.635)
Pseudo R^2	0.084	0.094	0.101	0.113	0.182	0.183
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.5: Probit regressions of Follow (complete table, Part 3/3)

Covariate block	Contribution to R^2	Share of increment (%)
Pre-treatment demographics	0.0176	8.5
Post-treatment demographics	0.0112	5.4
Technology attitudes	0.0122	5.9
Beliefs and norms	0.1634	79.2
General attitudes	0.0019	0.9
Treatment-and-framing R^2	0.0160	
Full-model R^2	0.2223	
Increment allocated above	0.2063	100.0

Table B.6: Grouped covariate contributions to explained variation in Follow

pre-treatment inference. The smaller difference in Initiate is stable after sequential inclusion of covariates. Predict and Advise remain statistically indistinguishable throughout.

	(1) Raw	(2) Minimal	(3) Pre-treatment	(4) Full	(5) Pair-clustered
Baseline: positive minus negative	0.027	0.030	0.037	0.015	0.037
	(0.045)	(0.045)	(0.045)	(0.035)	(0.044)
Predict: positive minus negative	0.664***	0.662***	0.661***	0.444***	0.661***
	(0.037)	(0.037)	(0.037)	(0.042)	(0.037)
Advise: positive minus negative	0.674***	0.677***	0.675***	0.475***	0.675***
	(0.037)	(0.037)	(0.037)	(0.042)	(0.036)
Initiate: positive minus negative	0.502***	0.503***	0.510***	0.309***	0.510***
	(0.043)	(0.043)	(0.043)	(0.043)	(0.042)
p(Predict = Advise)	0.842	0.778	0.786	0.589	0.784
p(Predict = Initiate)	0.005	0.005	0.008	0.022	0.007
p(Advise = Initiate)	0.003	0.002	0.004	0.005	0.003
p(symmetry)	0.757	0.756	0.831	0.397	0.834
Observations	1,596	1,596	1,596	1,596	1,596

Note: Entries are positive-minus-negative assessment differences in trust probability, with standard errors in parentheses. The full specification includes post-treatment covariates and is reported for completeness; the pre-treatment specification is the preferred one. Pair-clustered inference uses the pair-level assignment group.

Table B.7: Trust responsiveness: complete specifications

B.6 Follow by AI Role and Classifier Assessment

Table B.8 reports a probit model of Follow with AI-role-by-assessment interactions. Because Follow codes agreement as trust after a positive assessment and as no trust after a negative assessment, its interaction coefficients are not themselves the directional symmetry test. The main text therefore tests symmetry directly using the underlying trust probabilities in Table 2.

Dep. Var.: Follow	(1)
Predict	1.320***
	(0.130)
Advise	1.467***
	(0.130)
Initiate	1.100***
	(0.126)
Positive assessment	1.182***
	(0.136)
Predict × Positive assessment	-0.747***
	(0.206)
Advise × Positive assessment	-1.030***
	(0.204)
Initiate × Positive assessment	-0.904***
	(0.195)
Natural trust	0.168**
	(0.071)
Constant	-0.636***
	(0.095)
Observations	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Probit regression of Follow on AI role, positive classifier assessment, their interactions, and framing. Robust standard errors are in parentheses. In Baseline, the assessment was generated but not shown to participants.

Table B.8: Follow by AI role and classifier assessment

B.7 Framing Robustness

A possible concern is that the neutral versus natural trust framing could interact with AI role. Table B.9 reports probit regressions of Follow and trust with role-by-frame interactions. The ordering Predict $\approx$ Advise $>$ Initiate is present in both frames. The aggregate interaction tests provide no evidence that the main role pattern is confined to one description of the game. We do not interpret this as exact equality of every assessment-specific effect across frames.

Dep. Var.:	(1) Follow	(2) Trust
Predict	1.075***	-0.543***
	(0.138)	(0.131)
Advise	0.851***	-0.748***
	(0.132)	(0.130)
Initiate	0.645***	-0.593***
	(0.129)	(0.131)
Natural trust	0.138	-0.098
	(0.126)	(0.134)
Predict $\times$ Natural trust	-0.201	0.004
	(0.194)	(0.184)
Advise $\times$ Natural trust	0.365*	0.141
	(0.199)	(0.183)
Initiate $\times$ Natural trust	0.099	0.048
	(0.186)	(0.184)
Constant	-0.100	0.643***
	(0.089)	(0.096)
Observations	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Probit regressions of Follow (column 1) and trust (column 2) on AI role, natural trust frame, and their interactions. The omitted frame is the neutral frame. Robust standard errors in parentheses.

Table B.9: AI role by language-frame robustness

B.8 Exploratory Comprehension-Error Robustness

All participants had to correct any erroneous comprehension response before proceeding, so these questions were not an exclusion criterion for the analysis sample. As a deliberately stringent exploratory check, Table B.10 compares the primary Follow regression with the same specification for participants who recorded no mismatch on any comprehension attempt. The counters accumulate errors across attempts and, for the multiple-response question, across individual checkboxes. The restricted sample is therefore a zero-mismatch subset rather than a comprehension-pass sample. In total, 63.4% of participants recorded at least one mismatch before correction. Restricting the analysis to the 584 participants with no recorded mismatch leaves the treatment pattern qualitatively unchanged.

Dep. Var.: Follow	(1) Primary	(2) Zero mismatch
Predict	0.976***	0.912***
	(0.098)	(0.154)
Advise	1.009***	0.935***
	(0.097)	(0.157)
Initiate	0.693***	0.756***
	(0.093)	(0.159)
Natural trust	0.196***	0.283**
	(0.069)	(0.113)
Constant	-0.129*	-0.166
	(0.072)	(0.101)
Observations	1,596	584

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Probit regressions of Follow. Column 1 uses the analysis sample. Column 2 uses the subset of participants with no recorded failed attempts across any comprehension question. All participants had to correct errors before proceeding. Robust standard errors in parentheses.

Table B.10: Follow: exploratory zero-mismatch robustness

B.9 Beliefs and Social-Appropriateness Norms

After completing the trust game, participants reported their beliefs about the behaviour of others and their social-appropriateness judgements. Table B.11 presents the means by treatment. Kruskal–Wallis tests do not reject equality across treatments for beliefs about the share of Participants A trusting ($p = 0.912$) or the share of Participants B returning ($p = 0.033$). These post-choice measures are descriptive secondary outcomes and are not used to identify mediation. Perceived AI accuracy is also stable across the four treatments ($p = 0.981$).

B.10 Returning Behaviour

Table B.12 applies the same sequential specification structure used for Follow to Participant B's return decision. Column (2) is the pre-treatment benchmark. Columns (3)–(6) then add post-treatment covariate blocks, and column (7) repeats the pre-treatment model with pair-clustered inference. Participants made this decision after their own Participant A decision in the same treatment as their counterpart, but were not told which assessment their counterpart had received. The assignment can therefore affect returning only through participants' own experience of the AI role, not through information about the counterpart's assessment.

Return rates are 69.6% in Baseline, 72.2% in Predict, 66.0% in Advise, and 65.9% in Initiate. In the pre-treatment benchmark, Predict differs from Advise ($p = 0.046$) and Initiate ($p = 0.050$). Pair-clustered inference yields $p = 0.043$ and $p = 0.045$. The corresponding full-model values are $p = 0.063$ and $p = 0.101$, while Advise and Initiate remain nearly identical throughout. This secondary pattern is therefore not stable after sequential inclusion of covariates and receives no mechanism interpretation. It may reflect chance composition under adaptive pair-level assignment or features of the strategy-method sequence.

	Baseline	Predict	Advise	Initiate	Total
Belief that Participants A send (0-100)	55.78 (21.96)	55.24 (27.05)	54.50 (26.88)	54.55 (24.90)	55.02 (25.27)
Belief that Participants B return (0-100)	52.37 (22.14)	55.78 (25.67)	52.01 (24.68)	52.28 (23.75)	53.11 (24.13)
Appropriateness of Participant A sending (1-4)	3.359 (0.634)	3.323 (0.746)	3.280 (0.690)	3.275 (0.662)	3.310 (0.684)
Appropriateness of Participant A not sending (1-4)	2.221 (0.788)	2.361 (0.857)	2.280 (0.772)	2.293 (0.733)	2.289 (0.789)
Appropriateness of Participant B returning (1-4)	3.477 (0.664)	3.521 (0.625)	3.459 (0.643)	3.422 (0.638)	3.470 (0.643)
Appropriateness of Participant B not returning (1-4)	2.048 (0.875)	2.035 (0.879)	2.060 (0.829)	2.030 (0.789)	2.043 (0.843)
Perceived AI accuracy (1-5)	3.573 (0.780)	3.576 (0.779)	3.563 (0.848)	3.566 (0.835)	3.570 (0.810)

Note: Means (standard deviations in parentheses) by treatment. Beliefs are perceived shares of Participants A sending and of Participants B returning (0–100). Appropriateness items are rated on a 1–4 scale. Perceived AI accuracy ranges from 1 (very accurate) to 5 (random).

Table B.11: Beliefs and social-appropriateness norms by treatment

Dep. Var.: Returning	(1) Minimal	(2) Pre-treatment	(3) Post-treatment	(4) Technology	(5) Beliefs and norms	(6) Full	(7) Pair-clustered
Predict	0.076	0.099	0.095	0.100	0.096	0.113	0.099
	(0.094)	(0.095)	(0.095)	(0.096)	(0.113)	(0.114)	(0.095)
Advise	-0.102	-0.089	-0.091	-0.091	-0.124	-0.108	-0.089
	(0.093)	(0.093)	(0.094)	(0.095)	(0.109)	(0.109)	(0.094)
Initiate	-0.105	-0.086	-0.090	-0.090	-0.120	-0.089	-0.086
	(0.093)	(0.094)	(0.094)	(0.095)	(0.115)	(0.117)	(0.094)
Natural trust	0.285***	0.289***	0.285***	0.285***	0.346***	0.327***	0.289***
	(0.066)	(0.067)	(0.067)	(0.068)	(0.081)	(0.082)	(0.066)
p(Predict = Advise)	0.057	0.046	0.050	0.046	0.061	0.063	0.043
p(Predict = Initiate)	0.053	0.050	0.051	0.047	0.075	0.101	0.045
p(Advise = Initiate)	0.969	0.980	0.996	0.993	0.975	0.874	0.979
Pseudo R^2	0.012	0.025	0.028	0.042	0.292	0.308	0.025
Observations	1,596	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; pair-clustered inference in final column. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Probit regressions of Participant B's return decision. Column (2) is the pre-treatment benchmark. Columns (3)–(6) sequentially add post-treatment demographics, technology attitudes, beliefs and norms, and general attitudes. Column (7) repeats column (2) with standard errors clustered by pair-level assignment group.

Table B.12: Returning behaviour across specifications

Tables B.13–B.15 report all coefficient estimates from columns (1)–(6).

B.11 AI Prediction and Pre-Survey Demographics

The classifier generated its assessment from counterpart characteristics, so the assessment was not independently randomised. Table B.16 reports a descriptive probit of the AI assessment (1 = positive / trustworthy) on the available pre-survey demographics and framing. A joint test of all demographic regressors yields $p = 0.826$. Together with balance in classifier assessments across randomly assigned treatments, this supports comparability of the role-by-assessment patterns. It does not turn the assessment itself into an experimental treatment.

B.12 Classifier Assessments and Counterpart Behaviour

The classifier's disclosed accuracy of 89.86% refers to held-out performance on the training studies listed in Table A.1. In the experimental sample, its assessments had little discriminatory power. Among the 1,484 participants whose matched counterpart recorded a strategy-method return decision, counterparts returned 14 EP in 69.7% of cases after a positive assessment and in 66.6% of cases after a negative assessment. The 3.1 percentage-point difference is not statistically significant ($p = 0.200$, linear probability model with robust standard errors), and the assessment matched the counterpart's decision in 48.9% of cases, reflecting the high return rate in this population combined with a minority of positive assessments. This limited transfer from earlier laboratory and online studies to the present UK sample does not affect the treatment comparisons. Participants did not observe counterpart characteristics, received the same disclosed accuracy in all three AI treatments, and obtained no feedback before completing the questionnaire. The assessment and its relation to counterpart behaviour were identical across roles. Because participants could not detect classifier errors, differences in following across roles reflect the weight placed on the same information rather than differential error detection. For the same reason, we draw no conclusions about whether following the assessment improved participants' earnings.

Dep. Var.: Returning	(1)	(2)	(3)	(4)	(5)	(6)
Predict	0.076	0.099	0.095	0.100	0.096	0.113
	(0.094)	(0.095)	(0.095)	(0.096)	(0.113)	(0.114)
Advise	-0.102	-0.089	-0.091	-0.091	-0.124	-0.108
	(0.093)	(0.093)	(0.094)	(0.095)	(0.109)	(0.109)
Initiate	-0.105	-0.086	-0.090	-0.090	-0.120	-0.089
	(0.093)	(0.094)	(0.094)	(0.095)	(0.115)	(0.117)
Natural trust	0.285***	0.289***	0.285***	0.285***	0.346***	0.327***
	(0.066)	(0.067)	(0.067)	(0.068)	(0.081)	(0.082)
Age		0.007**	0.006*	0.005*	0.000	-0.002
		(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
Female		0.245***	0.228***	0.209***	0.289***	0.259***
		(0.067)	(0.070)	(0.074)	(0.084)	(0.086)
High school		-0.122	-0.098	-0.129	-0.189	-0.193
		(0.194)	(0.194)	(0.195)	(0.223)	(0.231)
Attended college		-0.130	-0.114	-0.127	-0.302	-0.332
		(0.203)	(0.204)	(0.206)	(0.237)	(0.244)
Undergraduate degree		-0.109	-0.078	-0.100	-0.274	-0.299
		(0.188)	(0.189)	(0.191)	(0.222)	(0.229)
Graduate degree or higher		-0.103	-0.078	-0.086	-0.310	-0.355
		(0.198)	(0.201)	(0.203)	(0.235)	(0.243)
Doctorate degree		-0.284	-0.234	-0.274	-0.583**	-0.629**
		(0.251)	(0.254)	(0.253)	(0.292)	(0.303)
Working		-0.063	-0.036	-0.005	-0.011	-0.014
		(0.091)	(0.098)	(0.099)	(0.114)	(0.118)
Student		-0.012	-0.026	0.004	0.010	-0.042
		(0.188)	(0.189)	(0.192)	(0.211)	(0.208)
Economics/Business		-0.124	-0.126	-0.107	-0.151	-0.126
administration background		(0.080)	(0.081)	(0.082)	(0.096)	(0.096)
Non-White		0.066	0.024	0.059	0.053	0.057
		(0.097)	(0.103)	(0.105)	(0.121)	(0.127)
£25.001 - £50.000			-0.116	-0.121	-0.085	-0.057
			(0.083)	(0.085)	(0.101)	(0.101)
p(Predict = Advise)	0.057	0.046	0.050	0.046	0.061	0.063
p(Predict = Initiate)	0.053	0.050	0.051	0.047	0.075	0.101
p(Advise = Initiate)	0.969	0.980	0.996	0.993	0.975	0.874
Pseudo R^2	0.012	0.025	0.028	0.042	0.292	0.308
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.13: Returning behaviour: complete specifications (Part 1/3)

Dep. Var.: Returning	(1)	(2)	(3)	(4)	(5)	(6)
£50.001 - £75.000			-0.057	-0.044	-0.028	0.014
			(0.115)	(0.118)	(0.139)	(0.139)
£75.001 - £100.000			0.126	0.116	0.291	0.300
			(0.199)	(0.200)	(0.215)	(0.218)
More than £100.000			-0.253	-0.251	-0.060	-0.010
			(0.210)	(0.212)	(0.236)	(0.233)
Born in England			0.016	0.008	-0.091	-0.101
			(0.081)	(0.082)	(0.093)	(0.094)
Religious			0.095	0.108	0.097	0.086
			(0.075)	(0.077)	(0.088)	(0.089)
Other			-0.018	-0.002	-0.050	-0.060
			(0.133)	(0.134)	(0.142)	(0.146)
Adults in household			-0.013	-0.020	-0.055*	-0.065**
			(0.029)	(0.029)	(0.032)	(0.032)
Children in household			0.026	0.027	0.020	0.012
			(0.035)	(0.036)	(0.041)	(0.041)
Experience with AI				-0.059	-0.155***	-0.142**
				(0.049)	(0.056)	(0.057)
Interaction with AI				0.063	-0.005	-0.010
				(0.044)	(0.050)	(0.051)
Profession with AI				0.146	0.066	0.026
				(0.116)	(0.138)	(0.139)
Attitude to AI				0.006	0.016	0.029
				(0.052)	(0.060)	(0.060)
Attitude to IT				0.033	-0.050	-0.056
				(0.049)	(0.054)	(0.055)
Locus of control				-0.106	-0.081	-0.086
				(0.072)	(0.084)	(0.085)
Reactance				-0.164***	-0.081	-0.014
				(0.052)	(0.059)	(0.062)
Belief in free will				-0.003	0.013	0.002
				(0.018)	(0.021)	(0.022)
p(Predict = Advise)	0.057	0.046	0.050	0.046	0.061	0.063
p(Predict = Initiate)	0.053	0.050	0.051	0.047	0.075	0.101
p(Advise = Initiate)	0.969	0.980	0.996	0.993	0.975	0.874
Pseudo R^2	0.012	0.025	0.028	0.042	0.292	0.308
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.14: Returning behaviour: complete specifications (Part 2/3)

Dep. Var.: Returning	(1)	(2)	(3)	(4)	(5)	(6)
(Would) consider prediction					-0.063	-0.075
					(0.047)	(0.047)
(Would) consider					0.102**	0.087*
recommendation					(0.048)	(0.048)
(Would) consider delegation					0.015	0.006
					(0.038)	(0.039)
Belief that Participants A					0.005**	0.004**
send (0-100)					(0.002)	(0.002)
Belief that Participants B					0.025***	0.025***
return (0-100)					(0.002)	(0.002)
Appropriateness of					0.262***	0.258***
Participant A sending (1-4)					(0.083)	(0.083)
Appropriateness of					-0.061	-0.032
Participant A not sending (1-4)					(0.075)	(0.076)
Appropriateness of					0.072	0.064
Participant B returning (1-4)					(0.088)	(0.089)
Appropriateness of					-0.356***	-0.345***
Participant B not returning (1-4)					(0.070)	(0.070)
Willingness to take risks						-0.022
						(0.019)
Trusting disposition						0.084*
						(0.047)
Altruism						0.003**
						(0.001)
Social image concerns						0.061***
						(0.017)
Constant	0.377***	0.149	0.186	0.965*	-0.132	-0.736
	(0.073)	(0.262)	(0.284)	(0.496)	(0.651)	(0.673)
p(Predict = Advise)	0.057	0.046	0.050	0.046	0.061	0.063
p(Predict = Initiate)	0.053	0.050	0.051	0.047	0.075	0.101
p(Advise = Initiate)	0.969	0.980	0.996	0.993	0.975	0.874
Pseudo R^2	0.012	0.025	0.028	0.042	0.292	0.308
Observations	1,596	1,596	1,596	1,596	1,596	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.15: Returning behaviour: complete specifications (Part 3/3)

Dep. Var.: AI Assessment (1=Positive)	(1)
Age	0.002
	(0.003)
Female	-0.060
	(0.066)
High school	0.233
	(0.181)
Attended college	0.275
	(0.190)
Undergraduate degree	0.141
	(0.176)
Graduate degree or higher	0.165
	(0.188)
Doctorate degree	0.160
	(0.240)
Non-White	-0.033
	(0.098)
£25.001 - £50.000	0.028
	(0.074)
£50.001 - £75.000	-0.030
	(0.105)
£75.001 - £100.000	0.093
	(0.181)
More than £100.000	-0.230
	(0.202)
Born in England	0.097
	(0.078)
Religious	0.018
	(0.071)
Other	-0.018
	(0.127)
Adults in household	-0.036
	(0.028)
Children in household	0.002
	(0.033)
Natural trust	0.119*
	(0.064)
Constant	-0.443*
	(0.245)
Pseudo R^2	0.007
Observations	1,596

Robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: Probit regression of the AI assessment (1 = positive / trustworthy) on pre-treatment demographics and framing. A joint test of all demographic variables yields $p = 0.826$.

Table B.16: Classifier assessment and pre-treatment demographics

C Pilot Study

The pilot study ran on Prolific in April 2024 with UK-based participants. Its purposes were to calibrate payoffs, timing, and the power calculation, and to collect return decisions from the participant pool of the main study for training the classifier (Table A.1). It was not preregistered. Table C.1 lists the differences from the main study. The most important is the fourth treatment, *Delegate*. Participants first chose whether to delegate the decision to the AI tool without seeing its decision. Those who delegated then saw the AI tool's decision, which was implemented without the option to override. Those who did not delegate made the decision themselves without seeing the assessment. In addition to the four treatments analysed here, the pilot included a variant of the advice treatment with an adaptive message style, whose return decisions enter the classifier's training data but which is not analysed in this paper.

	Pilot study (April 2024)	Main study
Endowment and multiplier	10 EP, $\times 3$	12 EP, $\times 2.5$
Return by Participant B	15 of 30 EP	14 of 30 EP
Trusting pays if $q \geq$	$2/3$	$6/7$
Disclosed classifier accuracy	83.07% (earlier classifier)	89.86%
Fourth treatment	Delegate (ex ante, binding)	Initiate (confirm or override)
Assessment shown in fourth treatment	Only after delegating	Always
Assignment	Treatment and frame by session	Pair-level adaptive randomisation
Completed participants	253	1,596

Table C.1: Design differences between the pilot study and the main study

The outcome of the Delegate treatment is the delegation choice. Of 61 participants, 29 (47.5%) delegated the decision to the AI tool, and 32 (52.5%) decided themselves, which meant deciding without the assessment. The delegation rate was 44.4% when the assessment that participants had not yet seen was negative and 50.0% when it was positive (Fisher's exact test, $p = 0.797$), as expected when the assessment is not visible at the time of the choice.

For completeness, Table C.2 describes how often final choices in the pilot coincided with the classifier assessment in each treatment.

Treatment	Final choice matches assessment (%)	N
Baseline	69.4	62
Predict	81.2	69
Advise	90.2	61
Delegate	65.6	61

Note: Entries are the share of participants who sent after a positive or did not send after a negative classifier assessment. In Baseline, the assessment was generated but not shown. In Predict and Advise, it was shown before the decision. In Delegate, it determined the final choice of the participants who delegated and was not shown to those who decided themselves. The Delegate entry therefore mixes the delegation choice with decisions taken without the assessment and does not correspond to Follow in the main study.

Table C.2: Pilot study: final choices and the classifier assessment

Stated preferences over AI roles replicate closely. Baseline participants in the pilot rated their willingness to use prediction at 3.45, advice at 3.19, and delegation at 2.48 on the same five-point scale, compared with 3.45, 3.37, and 2.39 in the main study. In the pilot, delegation is rated lower than both prediction and advice (paired t-tests, $p < 0.001$).

D Instructions

D.1 Reading guide

This appendix reproduces the instructions and decision screens as shown to participants. Text common to all participants is shown once. Participants were randomly assigned to one of the four AI-role treatments (Baseline, Predict, Advise, and Initiate) and one of the two language frames (neutral and natural trust). Square brackets mark wording that differed across treatments, frames, or classifier assessments; each variant is labelled, and the labels were not shown to participants. Interface elements are reported in italics.

D.2 Welcome and consent

Welcome

Welcome to this experiment. Thank you for participating.

You are taking part in an economic decision experiment. In this experiment, we apply a **strict no-deception policy**. This means that **all instructions provided to you** during this experiment are **truthful**.

Completing this experiment will take approximately **10–12 minutes**. Upon completion, you will receive £1.50. In addition, you will have the chance to earn "experimental points (EP)," depending on your decisions in the experiment. We will convert your EP to £ at the end of the experiment. The exchange rate is 8 EP = £1.00.

Please make sure to read the instructions carefully and to understand the tasks involved. If it is evident that you have not followed the instructions, you may not receive a payment.

The experiment will commence after you have read the Participation Information Sheet (PIS) and given your consent to participate.

[Next]

Please Confirm Your Consent

Participants received a link to the Participant Information Sheet for the research project "Economic Research on Human Interaction" (Cardiff Business School Research Ethics Committee Ref: 2322), led by Dr. Mengjie Wang (Cardiff University), and the following statements:

- I confirm that I have read the information sheet dated March 19, 2024, version 1 for the above research project.
- I confirm that I have understood the information sheet dated March 19, 2024, version 1 for the above research project and that I have had the opportunity to ask questions, and that these have been answered satisfactorily.
- I understand that my participation is voluntary, and I am free to withdraw at any time without giving a reason and without any adverse consequences.

- I understand that my pseudonymised personal information (demographic information) will be processed for the purposes explained to me, as set out in the information sheet, and that the researchers are not able to identify me personally in the data. I understand that such information will be held in accordance with all applicable data protection legislation and in strict confidence unless disclosure is required by law or professional obligation. I have been informed of my rights under data protection legislation and how I can raise any concerns.
- I understand who will have access to any personal information provided, how it will be managed, and what will happen to the data at the end of the research project.
- I understand that after the research project, de-identified data may be made publicly available via a data repository and may be used for purposes not related to this research project. I understand that it will not be possible to identify me from this data.
- I understand how the findings and results of the research project will be written up and published.
- I agree to take part in this research project.

Participants then had to tick: "I have read the information above and I agree to participate in this study."

D.3 Part 1: demographic questionnaire

Parts of the Experiment

This experiment consists of **two parts**.

In **Part 1**, you will be asked to answer some demographic questions. **No personal details** (e.g., your name or contact data) will be collected at any stage. Your responses will remain **completely confidential and anonymous** and will not be shared with other participants in this experiment.

Once you have finished Part 1, click the "next" button to ensure your responses are recorded.

In **Part 2**, you will be randomly matched with another participant. You and this other participant will stay anonymous to each other during and after the experiment. Together, you will complete a decision task.

You will begin by completing the demographic questions in Part 1. Once you have completed Part 1, you will receive more detailed instructions for Part 2.

Demographic questions

Participants reported age; gender identity (male, female, or other); continent of residence; education; employment status (multiple answers possible); whether they had studied or worked in business administration or economics; and ethnicity. They then proceeded to Part 2.

D.4 Part 2: common decision-task instructions

In this part, there are **two participants: Participant A and Participant B**.

You will **first** decide **in the role of Participant A**. Subsequently, you will decide **in the role of Participant B**.

The participant you are matched with will also first decide in the role of Participant A and subsequently decide in the role of Participant B.

At the end of this part, you will be **randomly assigned** either the **role of Participant A or Participant B**. The participant you are matched with will be assigned the opposite role. Your decision and the decision of the participant you are matched with in your randomly assigned roles will determine your payments.

D.4.1 Neutral frame

Participant A will initially be given 12 Experimental Points (EP), and Participant B will not be given any EP. First, Participant A will decide whether to **send** or **not to send** the 12 EP to Participant B. If Participant A chooses to **send**, Participant A's 12 EP will be sent to Participant B. Alternatively, Participant A can choose **not to send** the 12 EP to Participant B so that Participant A will keep the 12 EP.

If Participant A chooses to **send**, each EP will be multiplied by 2.5. Thus, Participant B will receive 30 EP. Then, Participant B decides whether or not to **return 14 EP** to Participant A. If Participant B chooses to **return**, Participant A will receive 14 EP and Participant B will keep 16 EP. Alternatively, Participant B can choose **not to return** and keep the entire 30 EP. If Participant A chooses **not to send** and keeps the 12 EP, then Participant B has no decision to make.

Decisions	A	B
A sends; B returns	14 EP	16 EP
A sends; B does not return	0 EP	30 EP
A does not send	12 EP	0 EP

D.4.2 Natural trust frame

Participant A will initially be given 12 Experimental Points (EP), and Participant B will not be given any EP. First, Participant A will have to decide whether to **trust** or to **distrust** Participant B. If Participant A chooses to **trust**, Participant A's 12 EP will be sent to Participant B. Alternatively, Participant A can choose to **distrust** Participant B. If Participant A chooses to distrust, Participant A will keep the 12 EP.

If Participant A chooses to **trust** Participant B, each EP will be multiplied by 2.5. Thus, Participant B will receive 30 EP. Participant B will then choose whether or not to be **trustworthy** to Participant A. If Participant B chooses to **be trustworthy**, 14 EP will be returned to Participant A. Therefore, Participant A will receive 14 EP and Participant B will keep 16 EP. Alternatively, Participant B can choose **not to be trustworthy** and keep the entire 30 EP. If Participant A chooses to **distrust** Participant B and keeps the 12 EP, then Participant B has no decision to make.

Decisions	A	B
A trusts; B is trustworthy	14 EP	16 EP
A trusts; B is not trustworthy	0 EP	30 EP
A distrusts	12 EP	0 EP

All participants were reminded: "Remember, the exchange rate is 8 EP = £1." They then answered two comprehension questions about the experiment, role assignment, payment, truthfulness, and exchange rate before continuing.

D.5 AI-tool introduction common to all treatments

An **artificial intelligence (AI) tool** was programmed [Baseline: for; Predict: to inform you in; Advise: to advise you in; Initiate: to support you in] this experiment. It was trained on roughly **16,000 decisions from previous participants in similar experiments** to **assess the decisions** of participants in the role of **Participant B based on answers to the questionnaire in Part 1**.

D.6 Treatment-specific instructions

Each participant saw one of the four AI-role treatments in one of the two language frames. The text for each treatment is reproduced once. Wording that varied with the language frame appears as [Neutral: . . . ; Natural trust: . . .], and wording that varied with the classifier assessment appears as [Positive: . . . ; Negative: . . .].

D.6.1 Baseline

Participants were told: "You will not have access to the AI tool." They then made Participant A's decision without receiving an AI assessment. The two options were [Neutral: "Send the 12 EP to Participant B." and "Do not send the 12 EP to Participant B."; Natural trust: "Trust Participant B." and "Do not trust Participant B."].

D.6.2 Predict

Before you decide, you will receive a **prediction** generated by the AI tool on Participant B's decision [Neutral: on whether to return the 14 EP or not in case you send the 12 EP; Natural trust: on whether to be trustworthy or not in case you trust].

The AI tool generates such a prediction by analysing data from numerous comparable tasks and incorporating demographic information provided by Participant B. This comprehensive analysis enables the system to make well-informed predictions based on patterns and trends observed across similar scenarios. The AI tool has an accuracy rate of 89.86% in its predictions, calculated by dividing the number of correct predictions by the total number of predictions. In other words, of every 100 predictions made by the AI tool, about 90 are correct.

The participant you are matched with will not receive any information about the prediction you have received and thus will not be aware of whether you incorporated the prediction of the AI tool into your decision.

The prediction read: "The AI tool predicts that Participant B will [Negative: not] [Neutral: return 14 EP to you if you send the 12 EP; Natural trust: be trustworthy if you trust]."

On the decision screen, participants saw the prediction again and chose between the two options listed for Baseline.

D.6.3 Advise

Before you decide, you will receive a **recommendation** generated by the AI tool regarding your decision [Neutral: on whether or not to send the 12 EP to Participant B; Natural trust: on whether or not to trust Participant B].

The AI tool generates such a recommendation by analysing data from numerous comparable tasks and incorporating demographic information provided by Participant B. This comprehensive analysis enables the system to make well-informed recommendations based on patterns and trends observed across similar scenarios. The AI tool has an accuracy rate of 89.86% in its recommendations, calculated by dividing the number of correct recommendations by the total number of recommendations. In other words, of every 100 recommendations made by the AI tool, about 90 are correct.

The participant you are matched with will not receive any information about the recommendation you have received and thus will not be aware of whether you incorporated the recommendation of the AI tool into your decision.

The recommendation read: "The AI tool recommends that you [Negative: do not] [Neutral: send the 12 EP to; Natural trust: trust] the other participant."

On the decision screen, participants saw the recommendation again and chose between the two options listed for Baseline.

D.6.4 Initiate

You will have the option to **delegate** your decision [Neutral: on whether or not to send the 12 EP to Participant B; Natural trust: on whether or not to trust Participant B] to the AI tool. After learning the AI tool's decision, you will have to decide whether to follow or override the AI tool's decision.

The AI tool generates its decision by analysing data from numerous comparable tasks and incorporating demographic information provided by Participant B. This comprehensive analysis enables the system to make informed decisions based on patterns and trends observed across similar scenarios. The AI tool has an accuracy rate of 89.86% in its decisions, calculated by dividing the number of correct decisions by the total number of decisions. In other words, of every 100 decisions made by the AI tool, about 90 are correct.

The participant you are matched with will not receive any information about whether you delegated your decision to the AI tool.

The AI tool's decision read: "The AI tool will [Negative: not] [Neutral: send the 12 EP to; Natural trust: trust] the other participant."

The decision screen stated: "Now you have to decide whether to delegate the decision to the AI tool or to make the decision yourself." The two buttons were [Neutral: "Stick with the AI tool's decision. (Send the 12 EP to Participant B.)" and "Override the AI tool's decision. (Do not send the 12 EP to Participant B.)"; Natural trust: "Stick with the AI tool's decision. (Trust Participant B.)" and "Override the AI tool's decision. (Do not trust Participant B.)"] after a positive assessment, and [Neutral: "Override the AI tool's decision. (Send the 12 EP to Participant B.)" and "Stick with the AI tool's decision. (Do not send the 12 EP to Participant B.)"; Natural trust: "Override the AI tool's decision. (Trust Participant B.)" and "Stick with the AI tool's decision. (Do not trust Participant B.)"] after a negative assessment.

D.7 AI-tool control questions

Participants in Predict, Advise, and Initiate answered which statement about the tool was correct. The correct option was: "Out of 100 cases, the AI tool's [Predict: prediction; Advise: recommendation; Initiate: decision] will be correct in about 90 cases." They then selected the two outcome combinations under which the AI tool's [Predict: prediction; Advise: recommendation; Initiate: decision] counted as correct. Participants could reopen the instructions while answering these questions.

D.8 Participant B decision

All participants subsequently made the decision in the role of Participant B using the strategy method: "Please make your decision in the role of Participant B now, assuming that **Participant A** [Neutral: **sent the 12 EP to you**; Natural trust: **trusted you**]." The two options were [Neutral: "Return 14 EP to Participant A." and "Do not return 14 EP to Participant A."; Natural trust: "Be trustworthy to Participant A." and "Do not be trustworthy to Participant A."].

D.9 Post-experimental questionnaire

Participants then completed the eleven-page final questionnaire. The social-appropriateness items were inspired by Krupka and Weber (2013). The questionnaire uses the scales for psychological reactance from Hong and Faedda (1996), trusting disposition from Gefen and Straub (2004) as adapted by Berger et al. (2021), personal innovativeness in information technology from Agarwal and Prasad (1998), locus of control

from Kovaleva (2012), risk tolerance from Dohmen et al. (2011), belief in free will from Harms et al. (2017), altruism from Falk et al. (2018), and attitudes towards AI from Sindermann et al. (2021). The remaining questions were designed for this study. Below we reproduce the wording and response scales shown to participants. As above, labelled square brackets display wording that varied with the language frame or treatment.

Background and household

- "Approximately how much did you personally earn, before taxes, during the past 12 months?" (No more than £25.000; £25.001–£50.000; £50.001–£75.000; £75.001–£100.000; More than £100.000.)
- "What is your country of birth?" (England; Northern Ireland; Scotland; Wales; Other [free text].)
- "Which of the following denominations best describes your religious identity?" (Roman Catholic; Protestant; Eastern Orthodox; Other Christian denomination; Jewish; Islam; Eastern religions; Other Non-Christian religions; Non-religious; None of the above.)
- "What is the number of adults in your household (including yourself)?" (1–30.)
- "What is the number of children in your household?" (0–30.)

Beliefs and social appropriateness

Participants answered: "Out of 100 participants in the role of Participant A in this experiment, how many do you think decided to [Neutral: send the 12 EP to; Natural trust: trust] Participant B?" and "Out of 100 participants in the role of Participant B in this experiment, how many do you think decided to [Neutral: return 14 EP to; Natural trust: be trustworthy to] Participant A?" (0–100.)

For each possible action, participants were asked to "state how appropriate or inappropriate [it is] for society, regardless of your own personal opinion". The actions were "[Neutral: Send the 12 EP to; Natural trust: Trust] Participant B", "Not [Neutral: send the 12 EP to; Natural trust: trust] Participant B", "[Neutral: Return 14 EP to; Natural trust: Be trustworthy to] Participant A", and "Not [Neutral: return 14 EP to; Natural trust: be trustworthy to] Participant A". Responses ran from 1 (totally inappropriate) to 4 (totally appropriate).

Experience with AI

- "Do you already have previous experience in using artificial intelligences?" (1, No, I have no experience at all, to 5, Yes, I have significant experience.)
- "How often (roughly) do you interact with generative artificial intelligence like chatGPT?" (Daily; Weekly; Once per month; Hardly ever; Never.)
- "Do you have a profession or education that is closely related to programming, robotics, or artificial intelligence?" (Yes; No.)

Psychological reactance

Participants indicated agreement with the following statements from 1 (strongly disagree) to 5 (strongly agree):

1. "I find contradicting others stimulating."

2. “It makes me angry when another person is held up as a role model for me to follow.”
3. “Regulations trigger a sense of resistance in me.”
4. “When something is prohibited, I usually think, ‘that’s exactly what I am going to do’.”
5. “I consider advice from others to be an intrusion.”
6. “I become frustrated when I am unable to make free and independent decisions.”
7. “It irritates me when someone points out things, which are obvious to me.”
8. “I become angry when my freedom of choice is restricted.”
9. “Advice and recommendations usually induce me to do just the opposite.”
10. “I resist the attempts of others to influence me.”
11. “When someone forces me to do something, I feel like doing the opposite.”

Trusting disposition

Using the same five-point agreement scale, participants evaluated: “I generally trust others”; “I generally have faith in others”; “I feel that others are generally well-meaning”; and “I feel that others are generally trustworthy.”

Personal innovativeness in information technology

Using the same five-point agreement scale, participants evaluated: “If I heard about a new information technology, I would look for ways to experiment with it”; “Among my peers, I am usually the first to try out new information technologies”; “In general, I am hesitant to try out new information technologies”; and “I like to experiment with new information technologies.” The page also contained the attention check: “Independent of what you think, please click on Neutral for this statement.”

Locus of control

Using the same five-point agreement scale, participants evaluated: “I am my own boss”; “If I work hard, I will succeed”; “Whether at work or in my private life: What I do is mainly determined by others”; and “Fate often gets in the way of my plans.”

Risk, free will, and social image

- “How do you see yourself: are you generally a person who is fully prepared to take risks or do you try to avoid taking risks?” (0, not at all willing to take risks, to 10, very willing to take risks.)
- “Please indicate to what extent you agree with the following statement: ‘I fully believe I have free will’.” (0, I fully disagree, to 10, I fully agree.)
- “Please indicate to what extent you agree with the following statement: ‘It is important for me not to be perceived as selfish’.” (0, I fully disagree, to 10, I fully agree.)

Altruism

Participants answered: "Imagine the following situation: Today you unexpectedly received £1,000. How much of this amount in Pounds would you donate to a good cause?" (£0–£1,000) and "How willing are you to give to good causes without expecting anything in return?" (0, not willing at all, to 10, very willing).

Attitudes towards AI

Participants indicated agreement from 1 (strongly disagree) to 5 (strongly agree) with: "I fear artificial intelligence"; "I trust artificial intelligence"; "Artificial intelligence will destroy humankind"; "Artificial intelligence will benefit humankind"; and "Artificial intelligence will cause many job losses." The page also contained the attention check: "Independent of your thoughts on Artificial Intelligence, please click on the most approving button for this statement."

Preferences over AI roles and perceived accuracy

Participants first read descriptions of the AI roles they had not experienced. All then indicated agreement from 1 (completely disagree) to 5 (completely agree) with one item for each role. For their experienced role, the item was: "I considered the prediction from the AI tool when making my decision whether to [Neutral: send the 12 EP to; Natural trust: trust] Participant B"; "I considered the recommendation from the AI tool when making my decision whether to [Neutral: send the 12 EP to; Natural trust: trust] Participant B"; or, in Initiate, "Had I chosen to stick with the decision the AI tool made, I would have been satisfied with that decision on my behalf." For roles not experienced, the corresponding items were: "I would have liked to receive a prediction on Participant B's decision on whether to [Neutral: return; Natural trust: be trustworthy] from the AI tool prior to my decision in the role of Participant A"; "I would have liked to receive a recommendation on my decision whether to [Neutral: send the 12 EP to; Natural trust: trust] Participant B from the AI tool prior to my decision in the role of Participant A"; and "I would have liked to delegate my decision on whether to [Neutral: send the 12 EP to; Natural trust: trust] Participant B to the AI tool prior to my decision in the role of Participant A." Baseline participants additionally evaluated "I would have used the AI tool."

Finally, after being reminded that the AI tool had an accuracy rate of 89.86%, participants answered: "How do you perceive the AI tool's accuracy of 89.86%?" (1, it is completely random, to 5, it is completely accurate.)

D.10 Payment

At the end, one of each matched pair's two role-contingent decisions was selected for payment. Participants whose partner had completed were shown the realised decisions, their fixed payment, decision-based payment, and total payment before returning to Prolific. If a partner had not yet completed, the participant was informed that the bonus would be processed later and was shown the complete payoff schedule.